



Orion-MC

Manual (Ventilation and Flap Control)

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1 General

1.1 Introduction

The Orion-MC is the communication interface between the main controller and the I/O modules in a greenhouse. The main controller sends the control commands to the Orion-MC. The Orion-MC converts these commands into control commands for the I/O modules. The Orion-MC communicates with the I/O modules via the CAN-Local data bus.

By default, the control occurs via digital inputs and outputs. The main controller can control the inputs of the Orion-MC directly with 24 Vac/dc.

With the Orion-MC, you can make motor groups to control the fans and flaps. You can assign two or more motors to each motor group. You can also use the Orion-MC to assign motor groups for different motor and flap controls, for example:

- ventilation
- air conditioning

1.2 The Type Plate

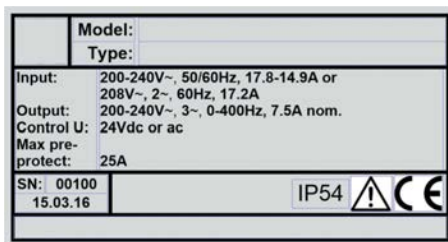


Fig. 1: Example of a type plate



This type plate is an example. The actual type plate on your product shows the correct information and technical data for that product.

The type plate shows important information and technical data about this product, such as:

- the model (related to the hardware configuration of the product)
- the type (related to the software and/or product settings)
- the technical data, such as the power supply, power consumption or pre-fuse value
- the symbols with regard to certifications and markings
- the serial number
- the date of manufacturing
- the contact information of the manufacturer

1.3 The Available Configurations

The Orion-MC is available in various configurations. The model indicates the type of hardware (including the power supply and/or the type of I/O board connected). The type indicates the type of software. The type plate shows the information for your product. This manual describes all the configurations.

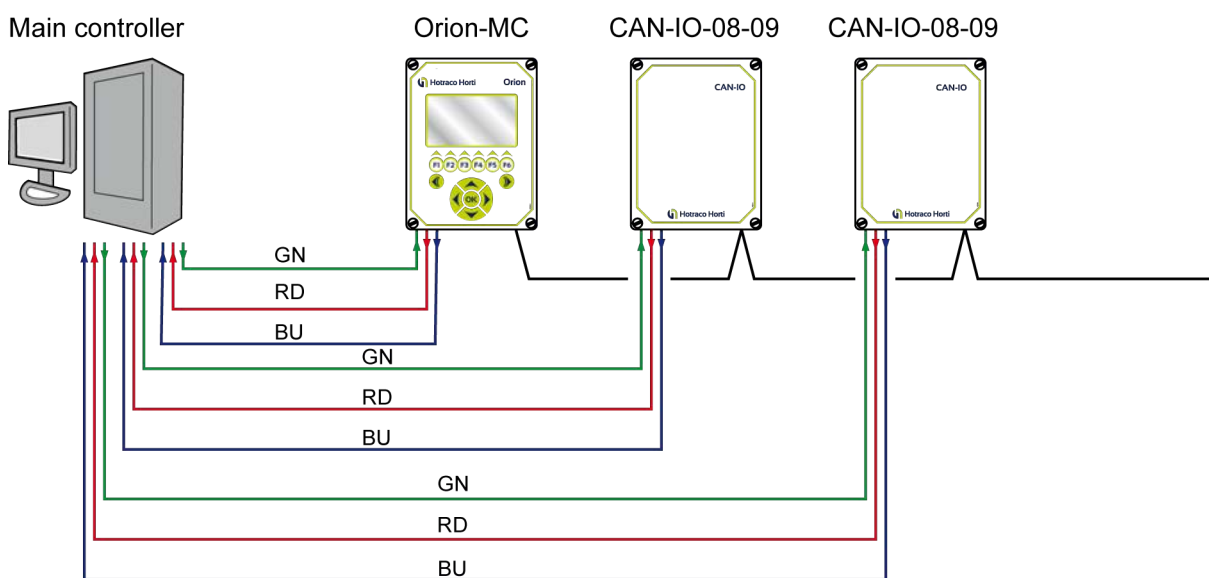
Tab. 1: Available configurations

Model	Type
- CAN-P-O - CAN-IO-8-9	Orion-MC

1.4 System Examples

1.4.1 Connection to the Main Controller

Digital Control (Orion-MC with CAN-IO-8-9)



GN Digital signal (control signal, ON/OFF, 24 Vdc)

RD Analog signal (feedback signal, 0-5 V or 0-20 mA)

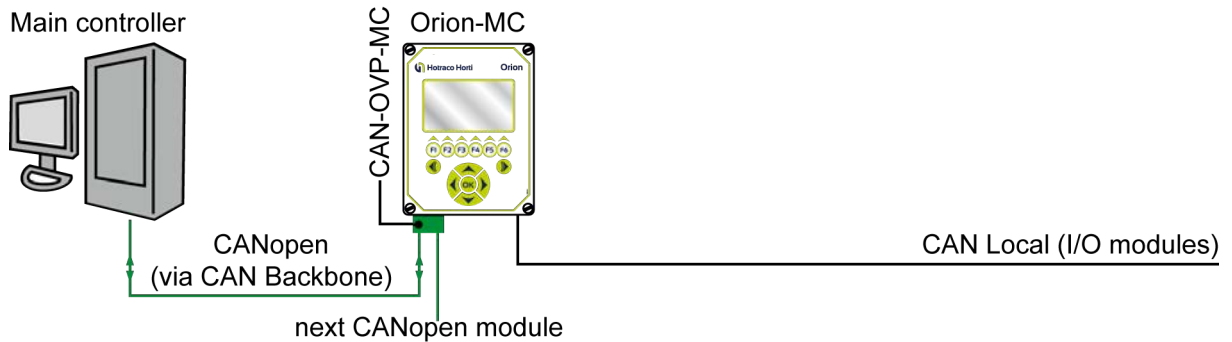
BU Digital signal (alarm messages, ON/OFF)

The main controller sends digital commands for each group to the Orion-MC (open, close, increase, decrease, for example). The virtual motor of the Orion-MC sends the set position as an analog signal back to the main controller. The main controller stops the sending of the digital command when the virtual motor has taken the required position. Digital feedback is given for alarms (the contact is disabled when an alarm occurs).



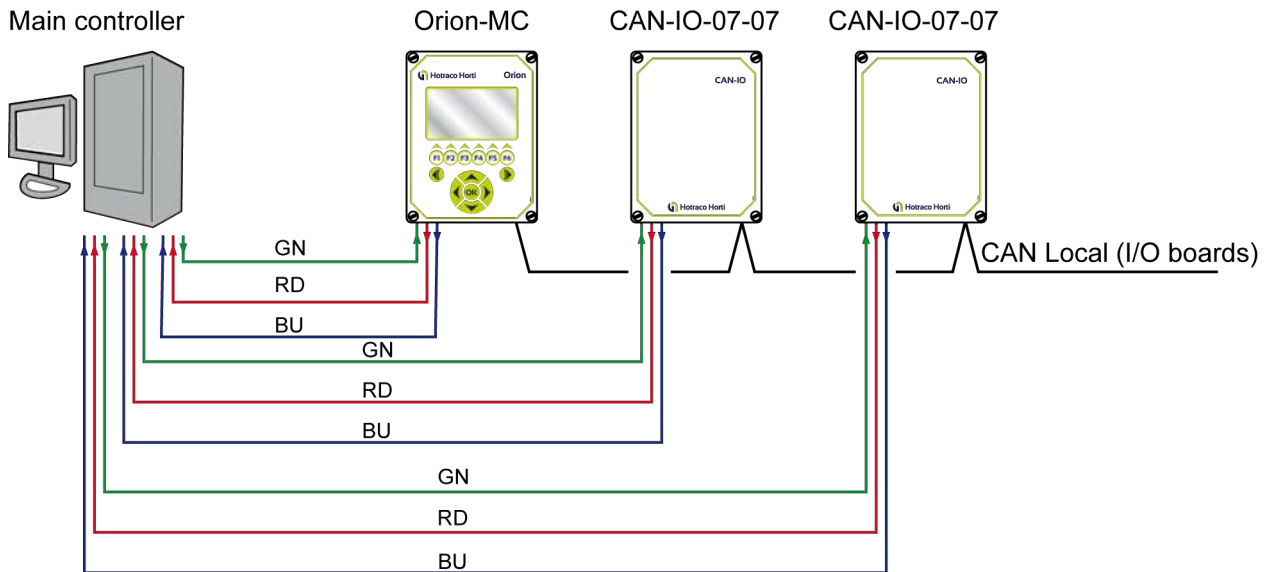
- You need two digital inputs and one analog output for each group.
- You need one digital output for each group for alarm messages.
- You can increase the number of inputs and outputs with additional CAN-IO-8-9 modules.

CANopen



With CANopen, the control signals and the feedback signals are sent via a CANopen data bus. To use the CANopen data bus, a CAN-OVP-MC module is necessary. Also, the installer must enable the CAN Backbone function during the installation procedure.

Analog Control (Orion-MC with CAN-IO-7-7)



- GN Analog signal (control signal, 0-5 V)
- RD Analog signal (feedback signal, 0-10 V)
- BU Digital signal (alarm messages, ON/OFF)

The main controller sends an analog command for each group to the Orion-MC that corresponds with the required position. The virtual motor of the Orion-MC sends the set position as an analog signal back to the main controller, so that the main controller receives the appropriate feedback. Digital feedback is given for alarms (the contact is disabled when an alarm occurs).



- You need one analog input and, if necessary, one analog output for each group.
- You need one digital output for each group for alarm messages.
- You can increase the number of inputs and outputs with additional CAN-IO-7-7 modules.

1.4.2 Flaps

Air Conditioning Cabinets



One RS485 bus can support maximum 31 addresses.

Example

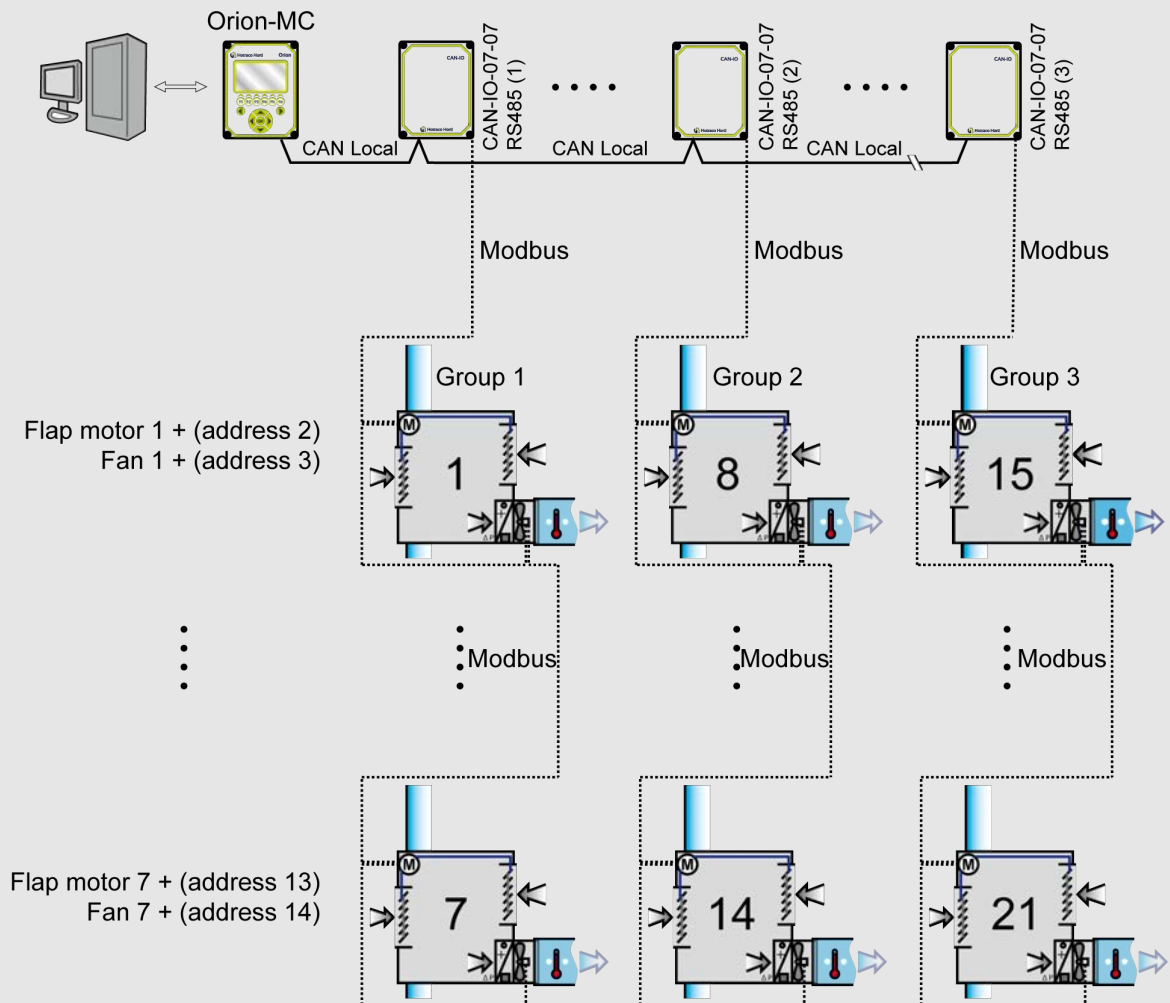
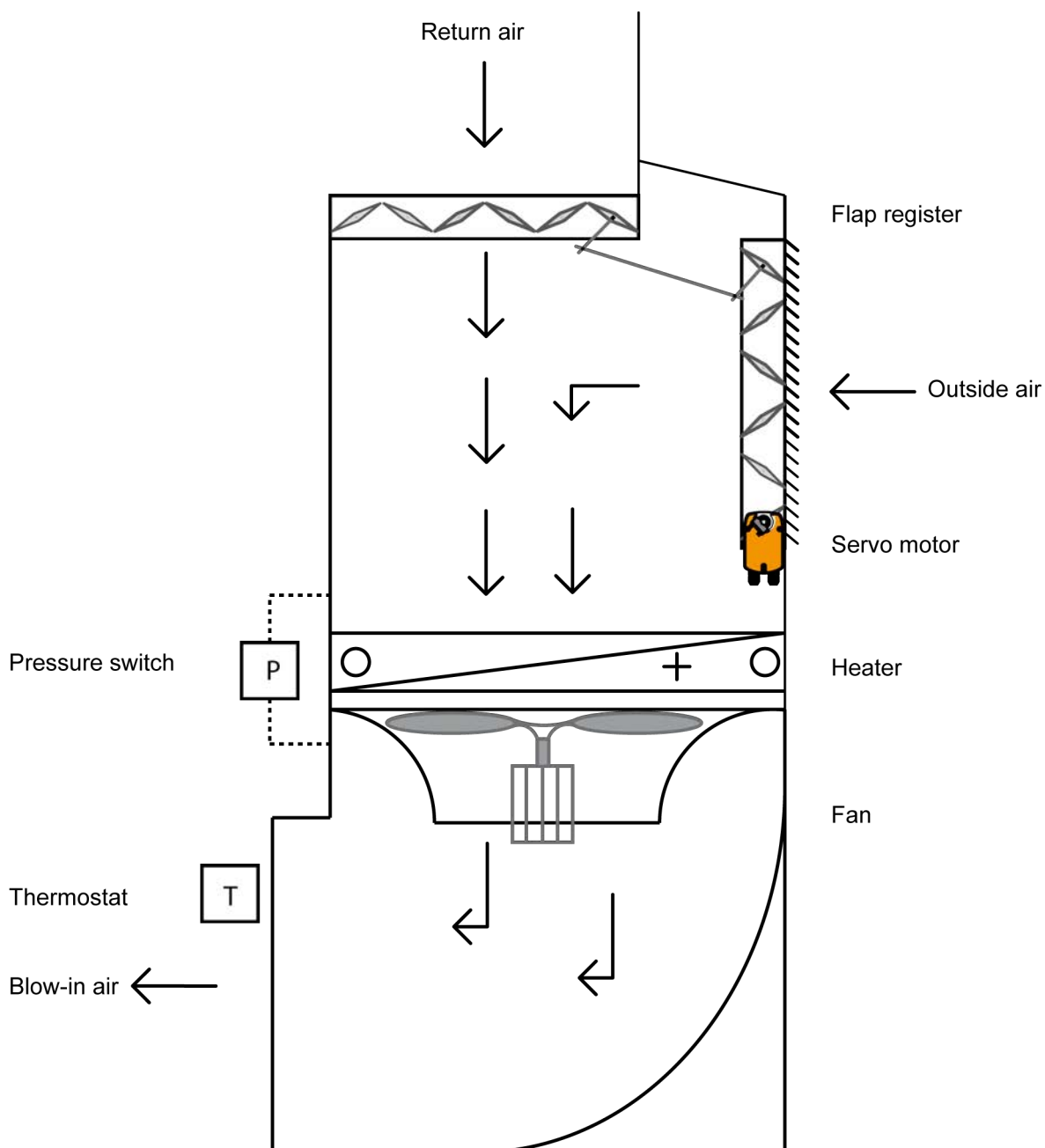


Fig. 2: CAN-IO-7-7 with Modbus

The main controller (not shown) determines the desired flap position of a group. The Orion-MC makes sure that the desired flap position is actually set.

The Orion-MC sends the desired flap position to the CAN-IO-7-7 modules through CAN Local. The I/O modules change the information into RS485 communication (Modbus), and send the information to the ventilation unit (flap motor and fan). Furthermore, the ventilation unit sends the actual flap position, the ventilation position, the blow-in temperature and the pressure difference of the heater back to the Orion-MC.

Active Ventilation System



1.4.3 Ventilation

CAN-IO-7-7 with EC Fans

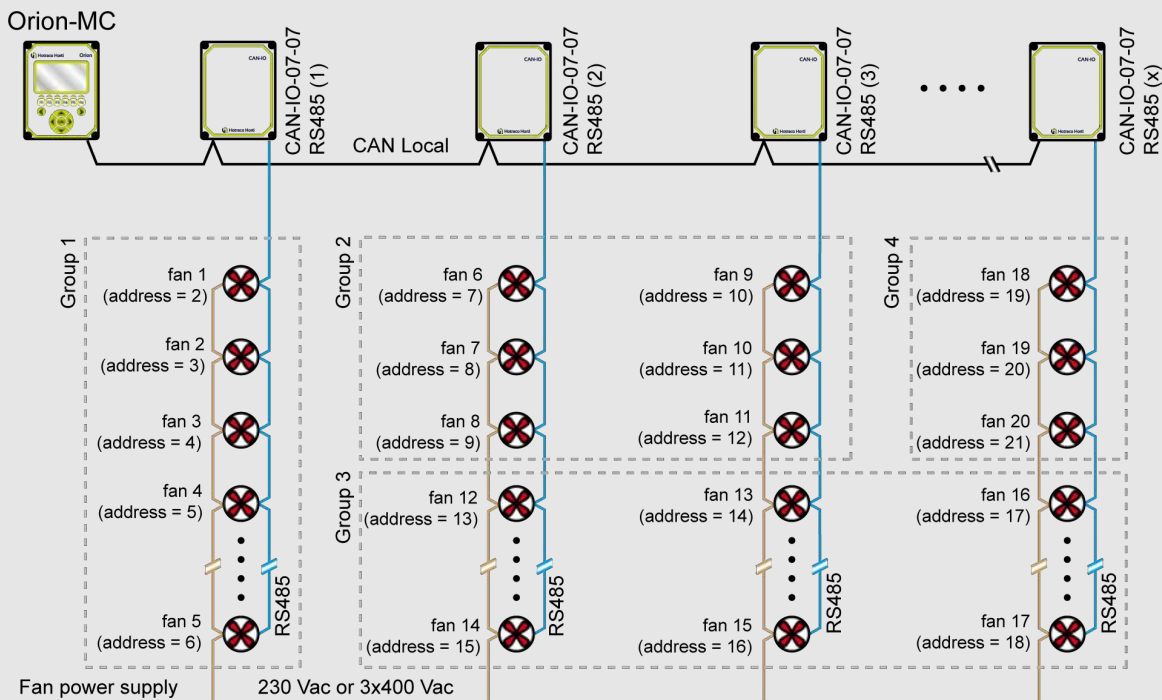


- One RS485 bus can support maximum 31 fans.
- The fans are numbered sequentially. In the example below:
 - group 1 has five fans (numbers 1 to 5)
 - group 2 has six fans (numbers 6 to 11)
 - group 3 has four fans (numbers 12 to 15)
 - group 4 has three fans (numbers 18 to 20)

It is not necessary that all the fans in a group are connected to the same bus.

- For the communication with the Orion-MC, you must set the address to the number of the fan +1. This makes it easier to replace a defective fan. The address of most new fans is 1. After you replaced the defective fan you must change the address of the new fan to the address of the original fan.

Example



The main controller (not shown) determines the desired ventilation position of a group. The Orion-MC makes sure that the desired ventilation position is actually set.

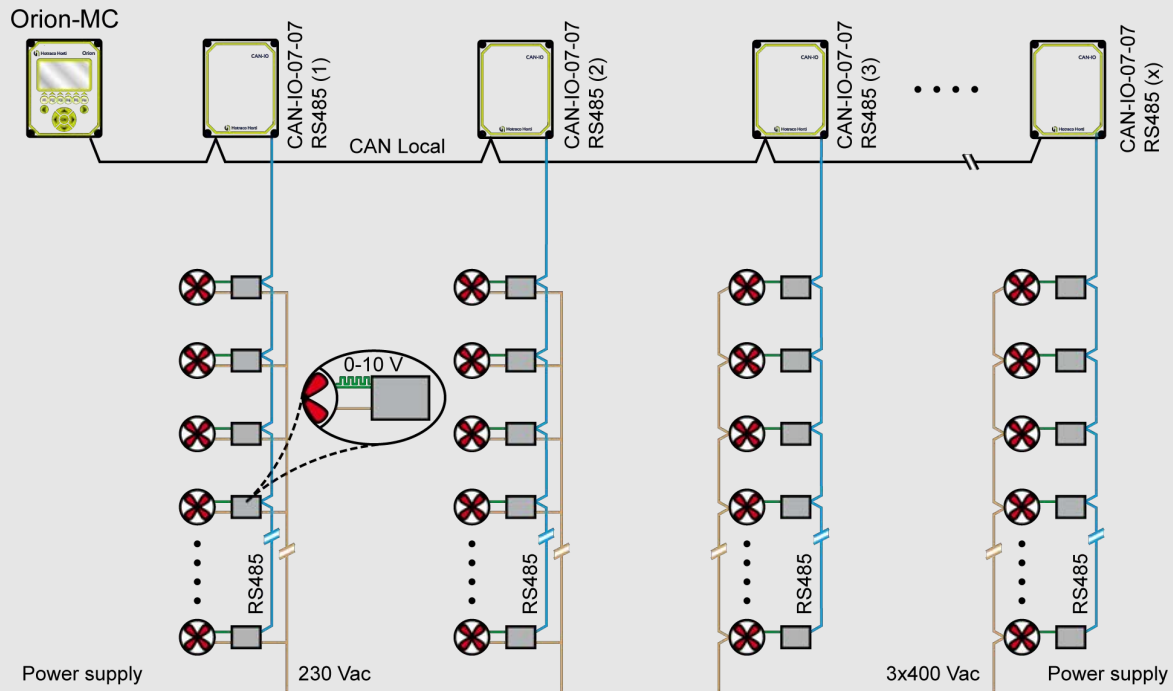
The Orion-MC sends the desired fan position to the CAN-IO-7-7 modules via the CAN Local connection. The I/O modules convert the information into RS485 communication (Modbus, for example). Then the EC fans take the desired speed automatically. The EC fans have an integral communication interface, and separate power supplies (230 Vac or 400 Vac). If necessary, the EC fans also send the actual fan position and alarm messages to the Orion-MC.

CAN-IO-7-7 with Separate EC Interface



- One RS485 bus can support maximum 31 fans.
- The classification of the groups is the same as for the EC fans.
- When you use 400 Vac fans, then each EC interface must have a separate power supply of 230 Vac.

Example



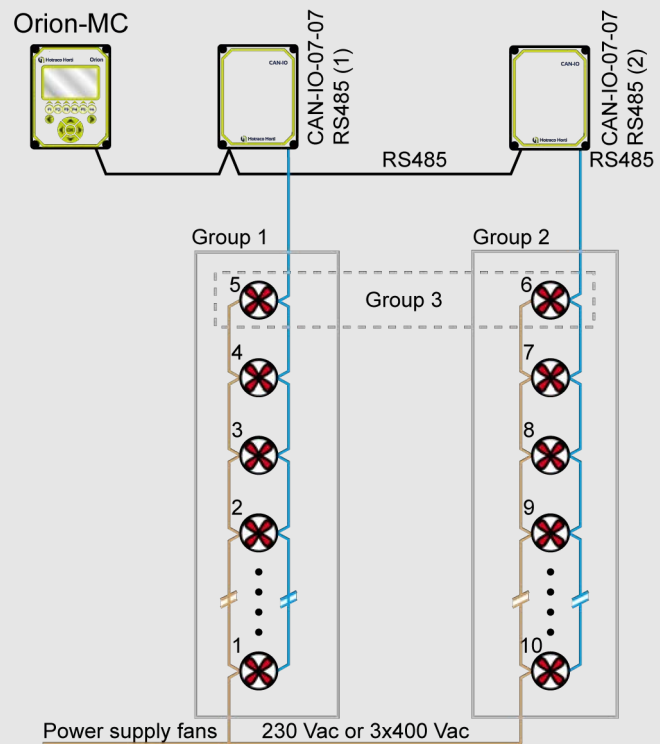
The main controller (not shown) determines the desired ventilation position of a group. The Orion-MC makes sure that the desired ventilation position is actually set.

The Orion-MC sends the desired fan position to the CAN-I/O-7-7 modules via the CAN Local connection. The I/O modules convert the information into RS485 communication (Modbus, for example). Then they send the information to the EC interface. The EC interface makes sure that the fans operate at the desired speed. If necessary, the EC interface also sends the actual fan position and alarm messages to the Orion-MC.

CAN-I/O-7-7 with EC Fans (Groups)

It is possible to switch fans in this group on or off, independent of the setpoint of the motor group. You must assign all the available fans to one or more digital inputs (maximum 12 digital inputs).

Example



- Group 1: Fans 1, 3, 5, 7, 9 (odd numbers)
- Group 2: Fans 2, 4, 6, 8, 10 (even numbers)
- Group 3: Fans 5 and 6

2 Safety

2.1 Introduction to Safety

- This manual contains the information required to install, operate and maintain this product. Read the information and instructions thoroughly before use.
- This product is covered by the warranty and liability regulations laid down in documents including the manufacturer's general terms of sale governing the contract concluded for the delivery of this product.

2.2 Description of the Signal Words

This document has safety-related and other important messages. Signal words and safety symbols describe the types of messages.



DANGER!

Indicates a hazardous situation which, if not avoided, will cause death or serious injury.

The safety symbol shows the type of hazard.



WARNING!

Indicates a hazardous situation which, if not avoided, could cause death or serious injury.

The safety symbol shows the type of hazard.



CAUTION!

Indicates a hazardous situation which, if not avoided, could cause minor or moderate injury.

The safety symbol shows the type of hazard.



NOTICE!

Indicates a hazardous situation which, if not avoided, could cause machine, property or product damage.



Indicates more information, suggestions and recommendations.

2.3 Description of the Safety Symbols

**WARNING!**

General warning.

**WARNING!**

Risk of injury from electrical voltage.

2.4 Personnel Qualification

Installer

Installers are all persons that do these actions with regard to the product:

- Installation
- Maintenance
- Troubleshooting

Installers must be qualified electricians. Each installer must have a profound professional experience and knowledge of:

- Electronic and electrical installations.
- The national and international regulations and legislation on electronic and electrical installations.
- Safety regulations.
- Possible dangers during the installation, maintenance and troubleshooting.
- Commissioning of the product.

Preferably, all installers have passed the product training of the manufacturer.

User

Users are all persons that operate this product. Users are only permitted to operate this product when:

- they have read and understood the instructions in this manual.
- they know the consequences of changes to settings.
- they know the possible dangers of incorrect use of this product.
- they know the possible dangers of the connected systems, especially systems that start automatically.
- during normal operation, they keep the cover of the housing closed, unless the manufacturer or this manual gives other instructions.

2.5 Safety Notes for the Installation

Personnel: ■ Installer

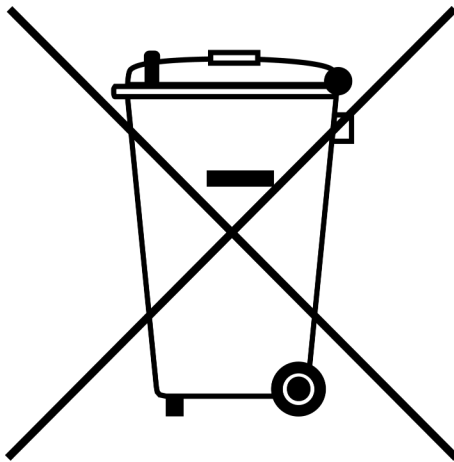
- The installation is an integral part of the product. Only professional installers are permitted to install this product.
- This product is under no circumstance a safety component.
- Install the product according to prevailing standards (BS, ANSI, ISO, DIN, NEN, for example).
- Before the installation, refer to the "Technical Specifications" section(s) in this manual.
- Secure the installation wiring to the product housing or cabinet to prevent short circuits.
- Connect the digital outputs (relay contacts) to the same electric potential. Do not mix mains and low-voltage signals.
- Always prevent ground loops. Be very careful with the analog outputs (0-5 V and 0-10 V). If possible, connect the components with galvanic isolation.
- Seal the cable glands with an acid-free sealant.
- Make sure that the system of which this product is part of, has an alarm device.
- Connect the alarm contacts of all connected systems to a central alarm unit. The manufacturer recommends the installation of an additional, independent alarm device (a minimum/maximum thermostat, for example).
- When motors or actuators are connected, install a suitable motor circuit protector and a lockable disconnect switch to disconnect the product from the mains supply.
- Install a lockable disconnect switch near the product. The disconnect switch must have a marking that clearly shows that this switch is the disconnecting device for this product.
- This product can be an integral part of a system with moving parts that start any time automatically. These moving parts could cause injury to persons. The moving parts must have guards or protective devices. The presence of persons near the moving parts is not permitted.
- When this product controls a moving or rotating system, Machinery Directive 2006/42/EC is applicable.

2.6 Safety Notes for the Usage

Personnel: ■ User

- Only persons who have read and understood the instructions in this manual are permitted to operate this product.
- Persons who operate or adjust this product must know the possible dangers of the connected systems.
- This product can control different systems. When the product is defective, it is unsafe! De-energize the connected systems until the product is repaired or replaced.
- This product is an electronic device. A malfunction or defect can always occur.
- Always keep the power ON to prevent condensation through cooling, unless the manufacturer or this manual gives other instructions.
- During normal operation, the cover of the housing must stay closed, unless the manufacturer or this manual gives other instructions.
- The system of which this product is part of, has an alarm device. At least once a day, do a check on the correct operation of this alarm device.
- This product can be an integral part of a system with moving parts that start any time automatically. These moving parts could cause injury to persons. The presence of persons near the moving parts is not permitted. At least once a day, do a check on the correct operation of the guards or protective devices.
- When this product controls a moving or rotating system, Machinery Directive 2006/42/EC is applicable.

2.7 End of Life (WEEE)



The crossed-out wheeled bin indicates a product that, at the end of its life cycle, must be collected separately (Directive 2012/19/EU on waste electrical and electronic equipment - WEEE). Separate collection is a precondition to ensure specific treatment and recycling of WEEE to get the chosen level of protection of human health and the environment in the Union. The black bar below the crossed-out wheeled bin indicates that the product was put on the market on or after August 13, 2005.

The European Union has systems in place for the separate collection of waste electrical and electronic products and batteries. Thus, return waste products, batteries and accumulators to the appropriate local authorities, or a local collection point/recycler. When you do not dispose of the product correctly, you may be fined.



Fig. 3: Crossed-out wheeled bin

3 Transport and Storage

- Check the product for transport damage.
- Check if the product is what you ordered.
In case of incorrect delivery or damage, contact the manufacturer.
- Store the product in a dry and clean environment:
 - Temperature: 0 °C ... 40 °C
 - Humidity: 20% ... 80%
- Store the product in such a way that it is protected from:
 - Humidity
 - Dust
 - Dirt
 - Direct sunlight
 - Aggressive, chemical substances
 - Mechanical vibrations

4 Installation

4.1 Installing the CAN-P-O

4.1.1 Technical Specifications CAN-P-O


NOTICE!

Different voltages can damage your product.

When an installation has two or more components that are fed separately, do a check if all the components can operate at the same mains voltage.


NOTICE!

Incorrect fuses can damage your product.

Always replace a defective fuse with a fuse of the same brand, type and fuse rating.

Electrical	
Type of mains supply	Fixed wiring (through a disconnect switch)
Operating area	Indoor, normal, dry (not suitable for wet areas)
Operating conditions	Continuous operation
Mains power supply	100-240 VAC
Frequency	50/60 Hz
Power consumption (full system)	Max. 0.75 A - 0.5 A
PCB fuse	T800mA (5 mm x 20 mm)
Pre-fuse	Max. 16 A
Insulation	Class 1 (IEC 61140)
Transient over-voltage, mains supply	Category II (IEC 60664-1)
Pollution degree	Category 2 (IEC 61010-1)

Mechanical	
Mobility	Fixed position (the product is mounted to the wall)
Operating temperature range	0-40 °C
Relative humidity	Max. 80%
Installation height	Max. 2000 m above sea level
Dimensions (H x W x D)	220 mm x 165 mm x 105 mm

Mechanical	
Housing	IP54, ABS
Weight	± 1 kg

Power supply	
Voltage 24 Vdc, ± 2.5%	Max. 800 mA

Communication	
CAN Local	Max. 500 m @ 100 kbps
CAN Backbone	Max. 500 m @ 125 kbps
RS232	Max. 15 m @ 38400 baud
USB	Max. 5 m @ 115 kbps
Ethernet	Max 100 m @ 10 Mbps

Cables	
CAN Local / CAN Backbone	Shielded twisted pair, 1x2x0.64 mm ² (CAN-BK cable)
RS232	MSK-9
USB	USB with USB-B mini plug and USB-A plug
Ethernet	UTP cat. 5e

Compliance	
EU Directives	See the Declaration of Conformity for this product

Wiring Diagram

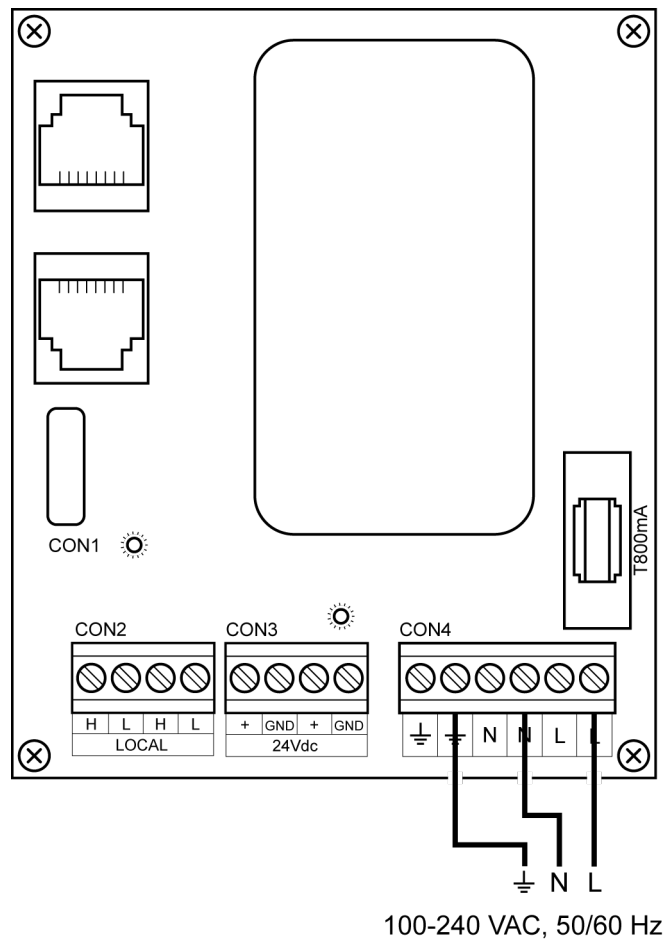


Fig. 4: CAN-P-O - Wiring diagram

Clean the Orion-MC



NOTICE!

When you clean the product incorrectly, you can damage it.

Detergents, solvents and running water can damage the product and the labels on it.

- Only use a soft cloth to clean the product.
- Only dampen the cloth with water.

4.1.2 Make the CAN Local Connection (CAN-P-O)

Personnel: Installer

Wiring Diagram

Terminal Connection

You can connect additional I/O boards to the CPU of the controller. The CAN Local bus provides the communication between the CPU board and the I/O boards. The communication with the first (internal) CAN-IO board occurs via the ribbon cable. When you connect additional I/O boards, you must connect them via the I/O board inside the controller.



Terminate the CAN Local connection with jumpers at the first and last communication node.

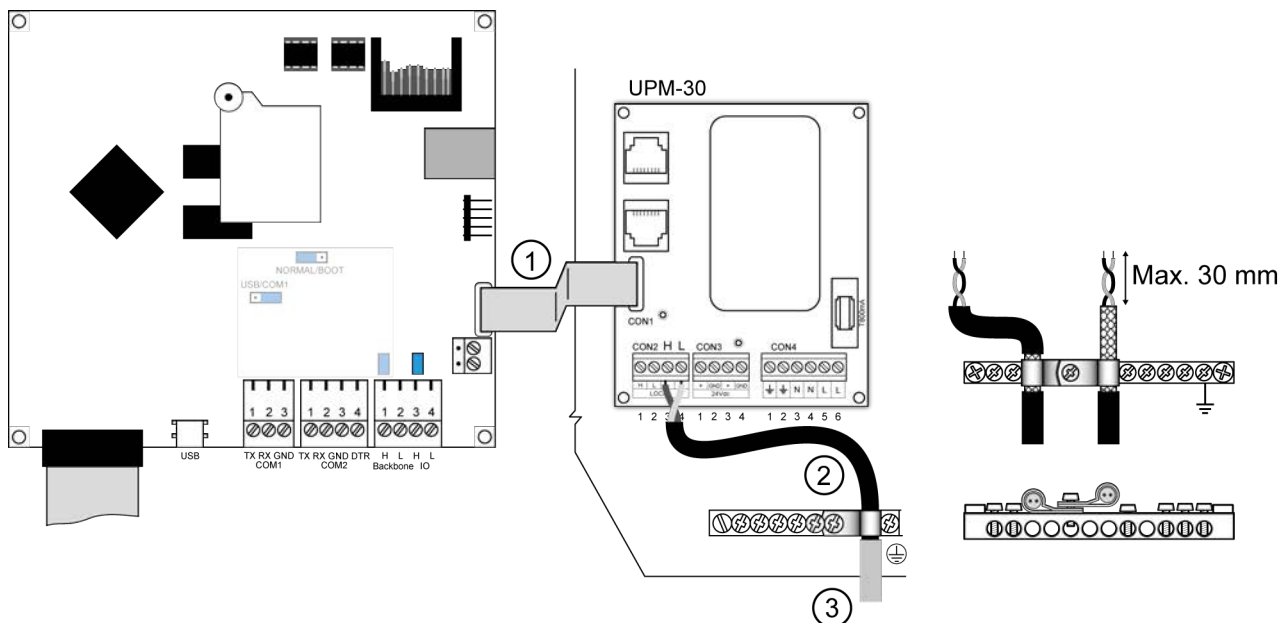


Fig. 5: CAN-P-O, CAN Local, wiring diagram

- ① Flat cable to CPU
- ② Shielded twisted pair CAN bus cable CAN-BK
1x2x0.64 mm² (max. 500 m)
- ③ CAN-P-O, CAN Local

Possible Connections Between Controllers

The CAN Local connection between controllers depends on the connection of the CPU. The examples below show the different possibilities.

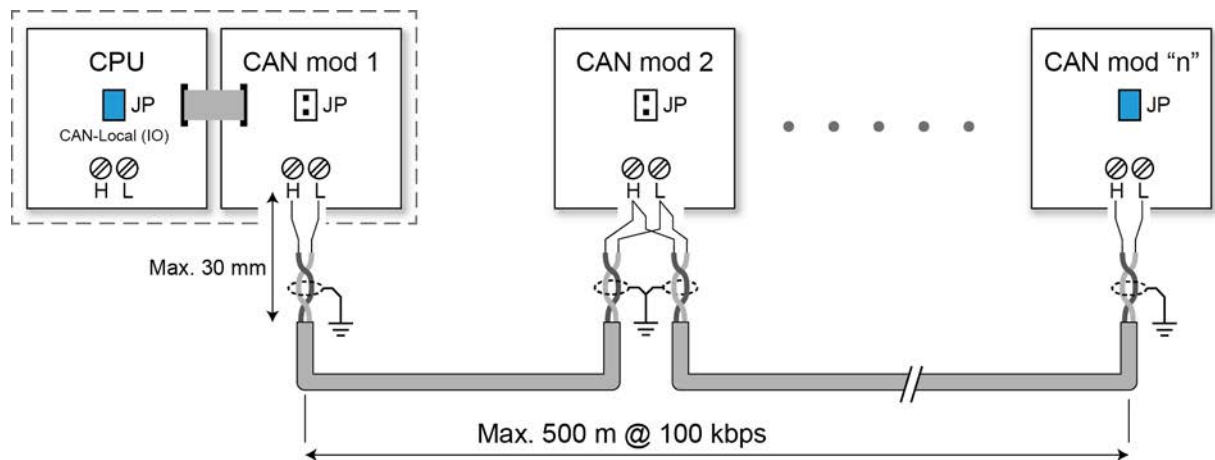


Fig. 6: CAN Local connection - Example 1

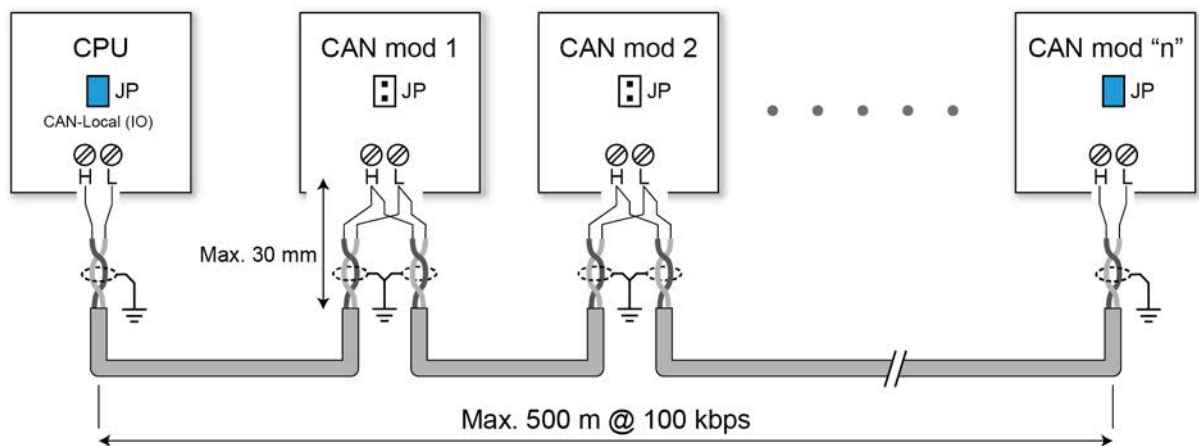


Fig. 7: CAN Local connection - Example 2

4.2 Installing the CAN-IO-8-9

4.2.1 Technical Specifications CAN-IO-8-9


WARNING!

Electric shock can cause serious injury.

This product can have more than one power source. Disconnect all the power sources before you access live or exposed parts of the product.


NOTICE!

Different voltages can damage your product.

When an installation has two or more components that are fed separately, do a check if all the components can operate at the same mains voltage.


NOTICE!

Incorrect fuses can damage your product.

Always replace a defective fuse with a fuse of the same brand, type and fuse rating.


NOTICE!

Ground loops can damage your product.

Always prevent ground loops when you install the product. If possible, provide galvanic isolation.

Electrical	
Type of mains supply	Fixed wiring (through a disconnect switch)
Operating area	Indoor, normal, dry (not suitable for wet areas)
Operating conditions	Continuous operation
Mains power supply	230 Vac $\pm$ 10%
Power supply 24 V	24 Vac $\pm$ 10%
Frequency	50/60 Hz
Power consumption	Max. 15 VA
PCB fuse	F100mA (5 mm x 20 mm)
Insulation	Class 1 (IEC 61140)
Transient over-voltage, mains supply	Category II (IEC 60664-1)
Pollution degree	Category 2 (IEC 61010-1)

Mechanical	
Mobility	Fixed position (the product is mounted to the wall)
Operating temperature range	0-40 °C
Relative humidity	Max. 80%
Installation height	Max. 2000 m above sea level
Dimensions (H x W x D)	220 mm x 165 mm x 105 mm
Housing	IP54, ABS
Weight	± 1 kg

Digital inputs (8)	
Open/close, higher/lower	NPN/PNP sensor, 12...24 Vac/dc, 8 mA, max. 10 Hz

Analog outputs (4)	
Feedback, 0-5 Vdc	Max. 1 mA
Feedback, 0-20 mA	Max. 10 Vdc

Digital outputs (4 + 1 alarm output)	
Relay output K1-K4	0.5 A, 230 Vac or 0.5 A, 24 Vac/dc
Alarm output	0.5 A, 24 Vac/dc

Power supply	
Voltage 24 Vdc -30% +10%	Max. 200 mA

Communication	
CAN Local	Max. 500 m @ 100 kbps
CAN Backbone	Max. 500 m @ 125 kbps

Cables	
CAN Local / CAN Backbone	Shielded twisted pair, 1x2x0.64 mm ² (CAN-BK cable)

Compliance

EU Directives

See the Declaration of Conformity for this product

4.2.2 Installation

Introduction

 Personnel: Installer

The CAN-IO-8-9 is an input/output module with:

- 8 digital inputs
- 4 analog outputs
- 4 digital outputs + 1 alarm output
- a CAN bus.

Setting the Address



WARNING!

Electric shock can cause serious injury.

Always disconnect the power supply to the product before you change the DIP switch settings.

The CAN-IO-8-9 communicates with external devices through the CAN Local bus. The table shows the DIP switch setting for each device.

#	SW1	#	SW1	#	SW1	#	SW1
1		5		9		13	
2		6		10		14	
3		7		11		15	
4		8		12		16	

1.  Switch off the power supply.
2.  Set the address with DIP switch SW1.
3.  Switch on the power supply. The new DIP switch setting is read.

Wiring Diagram

**WARNING!**

Electric shock can cause serious injury.

Always disconnect the power supply to the product before you do work on electrical parts inside the product. When this is not possible, be very careful that you do not touch live parts.

**WARNING!**

Electric shock can cause serious injury.

This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.

**NOTICE!**

Electrostatic discharge can damage your product.

All the electrical parts inside this product are susceptible to electrostatic discharge (ESD).

- Always disconnect the power supply to the product before you do work on electrical parts inside the product.
- Always wear an ESD wrist strap that is connected to the main earthing system.

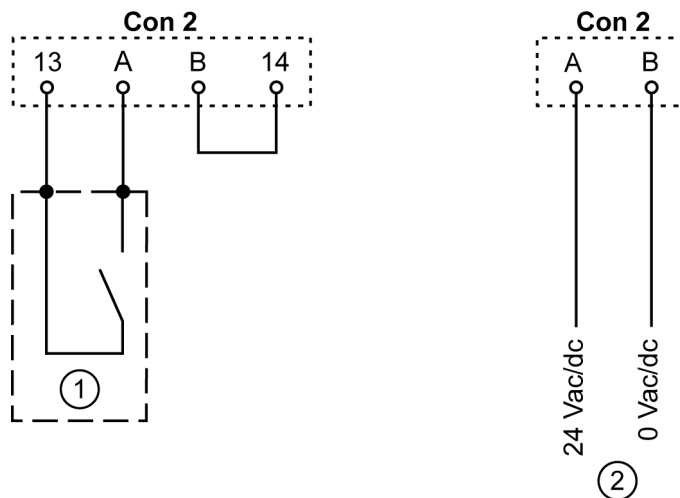
See IEC 61340-5-2 for more information on precautions with regard to ESD.

**NOTICE!**

Ground loops can damage your product.

Always prevent ground loops when you install the product. If possible, provide galvanic isolation.

Digital Inputs



- ① External contact
- ② External control

- ① NPN contact
- ② PNP contact
- ③ Voltage-free contact
- ④ Depending on the other connections to this group

Analog Outputs

Digital Outputs

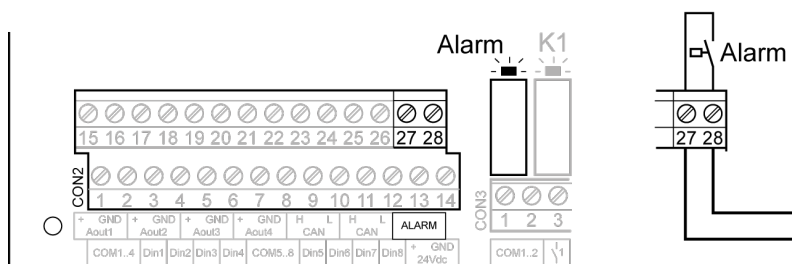


NOTICE!

A mix of voltages at the relay contacts can damage your product.

- Connect the digital outputs (relay contacts) to the same electric potential.
- Do not mix mains and low-voltage signals.

Alarm



Possible Connections between CAN Modules

The CAN Local connection between CAN modules depends on the connection of the CPU. The examples below show the different possibilities.

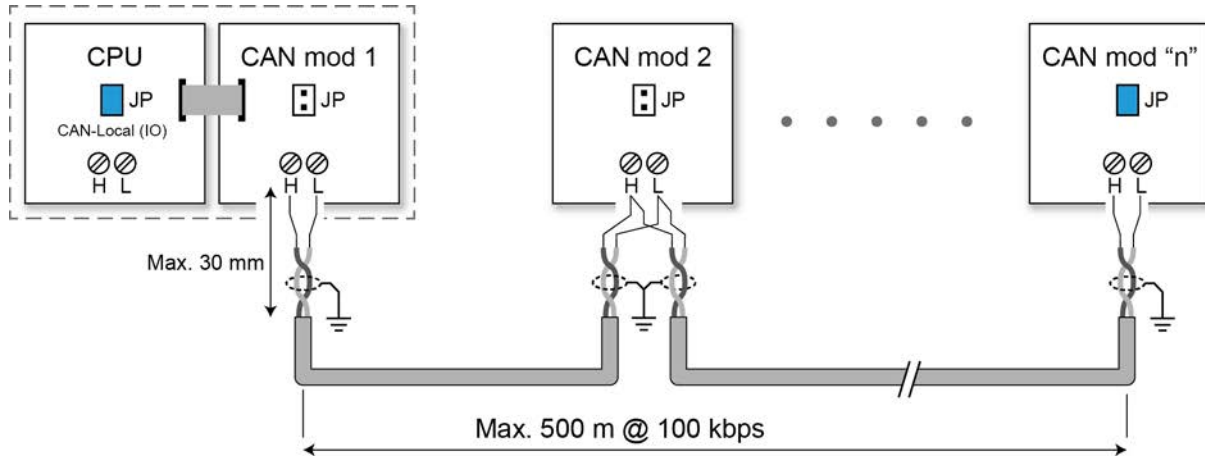


Fig. 9: CAN Local connection - Example 1

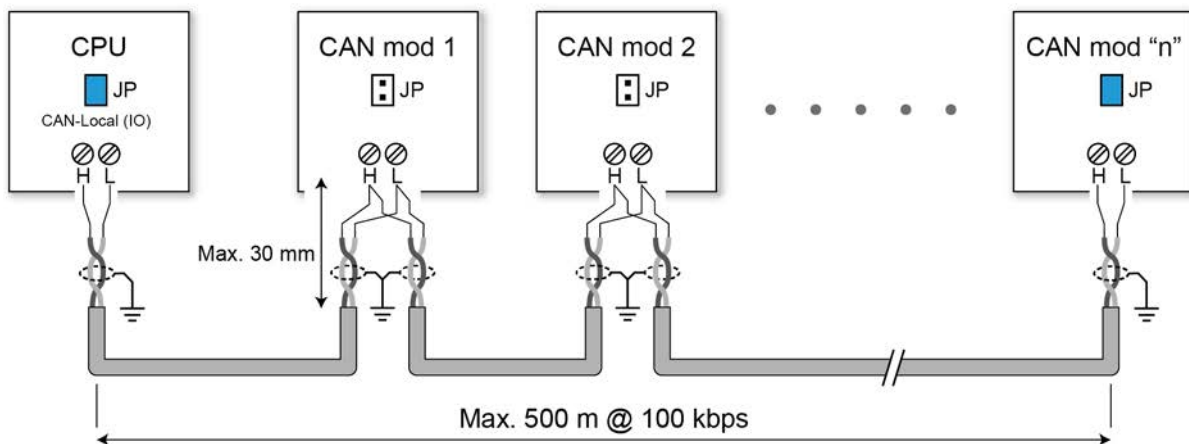


Fig. 10: CAN Local connection - Example 2



Terminate the CAN Local connection with jumpers at the first and last communication node.

4.3 Installing the CAN-IO-7-7

4.3.1 Technical Specifications CAN-IO-7-7


WARNING!

Electric shock can cause serious injury.

This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.


NOTICE!

Different voltages can damage your product.

When an installation has two or more components that are fed separately, do a check if all the components can operate at the same mains voltage.


NOTICE!

Incorrect fuses can damage your product.

Always replace a defective fuse with a fuse of the same brand, type and fuse rating.


NOTICE!

Ground loops can damage your product.

Always prevent ground loops when you install the product. If possible, provide galvanic isolation.

Electrical	
Types of mains supply	Fixed wiring
Operating area	Indoor, normal, dry (not suitable for wet areas)
Operating conditions	Continuous operation
Mains power supply	230 Vac $\pm$ 10%
Power supply 24 V	24 Vac $\pm$ 10%
Frequency	50/60 Hz
Power consumption	Max. 15 VA
PCB fuse	T100mA (5 mm x 20 mm)
Insulation	Class 1 (IEC 61140)

Electrical

Transient over-voltage, mains supply	Category II (IEC 60664-1)
Pollution degree	Category 2 (IEC 61010-1)

Mechanical

Mobility	Fixed position (the product is mounted to the wall)
Operating temperature range	0 °C ... 40 °C
Relative humidity	Max. 80%
Installation height	Max. 2000 m above sea level
Dimensions (H x W x D)	220 mm x 165 mm x 105 mm
Housing	IP54, ABS
Weight	± 1 kg

Analog inputs (4)

Temperature sensor (PT-1000)	-40 °C ... 100 °C ± 0.5 °C
Potentiometer	5 kΩ
0-20 mA	$R_i = 250 \Omega$
0-10 V	$R_i = 50 \text{ k}\Omega$

Digital inputs (3)

Contact / counter	NPN / PNP sensor, 12-24 Vac/dc, 8 mA, max. 10 Hz
-------------------	--

Analog outputs (2)

0-10 Vdc	0-10 V max. 1 mA
----------	------------------

Digital outputs (4 + 1 alarm output)

Relay output K1-K4	0.5 A, 230 Vac or 0.5 A, 24 Vac/dc
Alarm output	0.5 A, 24 Vac/dc

Power supply

Voltage 24 Vdc -30% +10%	Max. 300 mA
--------------------------	-------------

Communication	
CAN Local	Max. 500 m @ 100 kbps
CAN Backbone	Max. 500 m @ 125 kbps
Ethernet	TCP/IP, max 100 m @ 10 Mbps
RS485	Max. 1200 m (depends on the speed)

Cables	
CAN Local / CAN Backbone	Shielded twisted pair, 1x2x0.64 mm ² (CAN-BK cable)
RS485	Shielded twisted pair, 1x2x0.64 mm ² (RS485 cable)

Compliance	
EU Directives	See the Declaration of Conformity for this product

4.3.2 Installation

Introduction

Personnel: Installer

The CAN-IO-7-7 is an input/output module with:

- 4 analog inputs
- 3 digital inputs
- 2 analog outputs
- 4 digital outputs + 1 alarm output
- a CAN bus
- an RS485 bus.

Setting the Address



WARNING!

Electric shock can cause serious injury.

Always disconnect the power supply to the product before you change the DIP switch settings.

The table shows the DIP switch setting for each CAN-IO-7-7 (maximum 32 addresses).

#	SW1	#	SW1	#	SW1	#	SW1
1		9		17		25	
2		10		18		26	
3		11		19		27	
4		12		20		28	
5		13		21		29	
6		14		22		30	
7		15		23		31	
8		16		24		32	

1. ➤ Switch off the power supply.
2. ➤ Set the address with DIP switch SW1.
3. ➤ Switch on the power supply. The new DIP switch setting is read.

Wiring Diagram

**WARNING!**

Electric shock can cause serious injury.

Always disconnect the power supply to the product before you do work on electrical parts inside the product. When this is not possible, be very careful that you do not touch live parts.

**WARNING!**

Electric shock can cause serious injury.

This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.

**NOTICE!**

Different voltages can damage your product.

When an installation has two or more components that are fed separately, do a check if all the components can operate at the same mains voltage.

**NOTICE!**

Electrostatic discharge can damage your product.

All the electrical parts inside this product are susceptible to electrostatic discharge (ESD).

- Always disconnect the power supply to the product before you do work on electrical parts inside the product.
- Always wear an ESD wrist strap that is connected to the main earthing system.

See IEC 61340-5-2 for more information on precautions with regard to ESD.

**NOTICE!**

Ground loops can damage your product.

Always prevent ground loops when you install the product. If possible, provide galvanic isolation.



NOTICE!

A mix of voltages at the relay contacts can damage your product.

- Connect the digital outputs (relay contacts) to the same electric potential.
- Do not mix mains and low-voltage signals.



Terminating CAN Local

Always terminate the CAN Local connection at the first and last communication node. There, the CAN Local jumper must be closed. The CAN Local jumpers of the CAN modules in-between must be open.

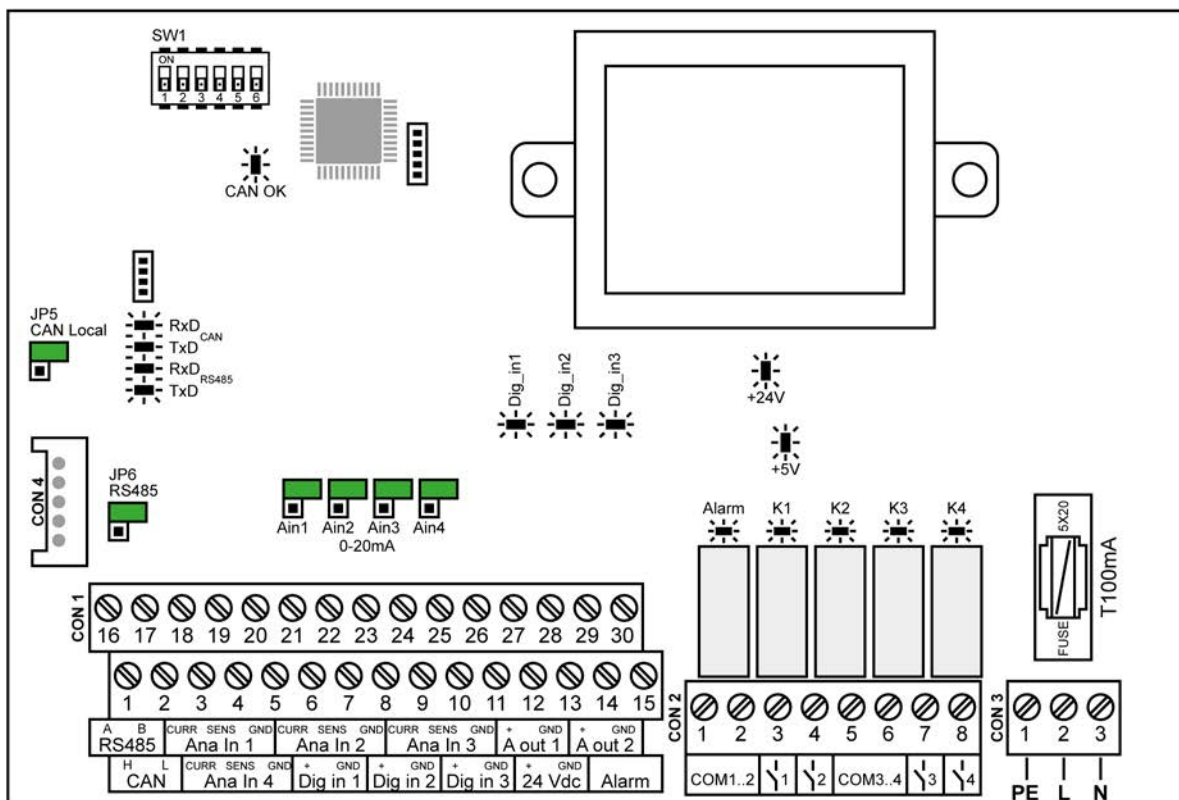


Fig. 11: CAN-IO-7-7 - Wiring diagram

Analog Inputs

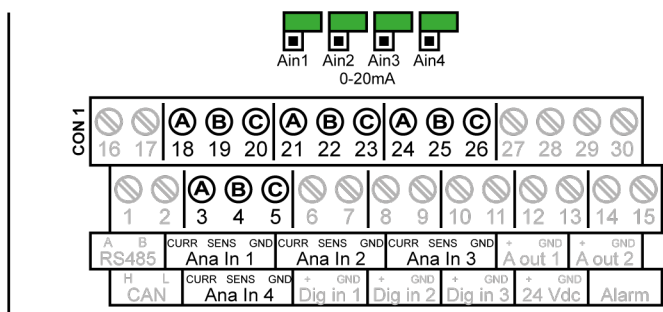


Fig. 12: CAN-IO-7-7 - Analog inputs - General

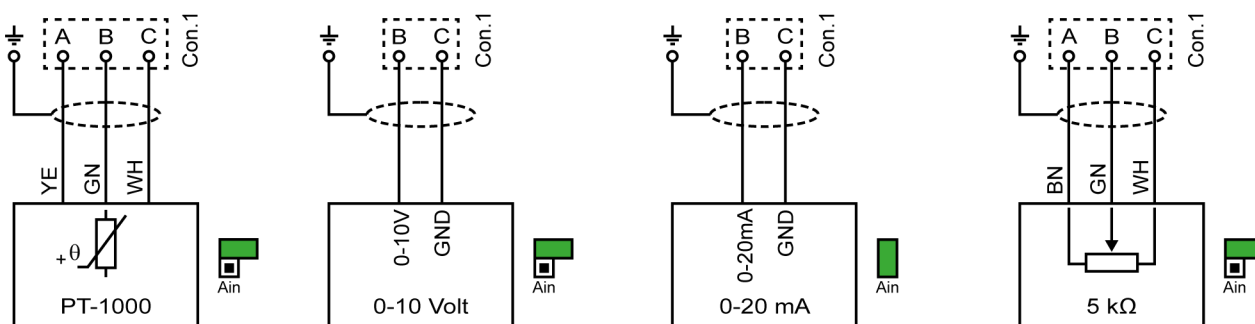


Fig. 13: CAN-IO-7-7 - Analog inputs - Jumper positions

Digital Inputs

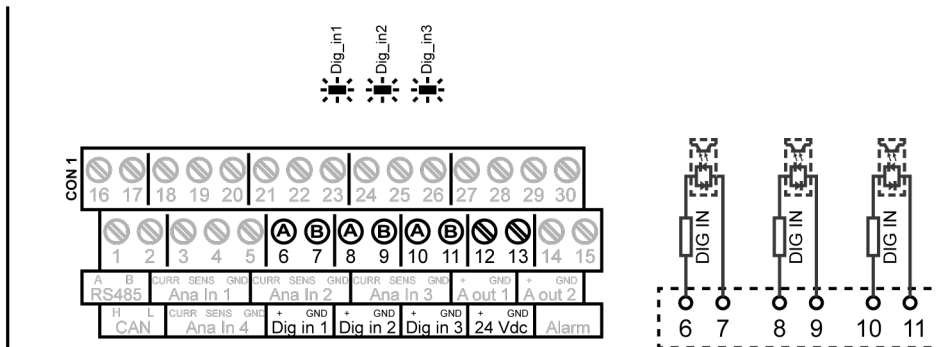


Fig. 14: CAN-IO-7-7 - Digital inputs - General

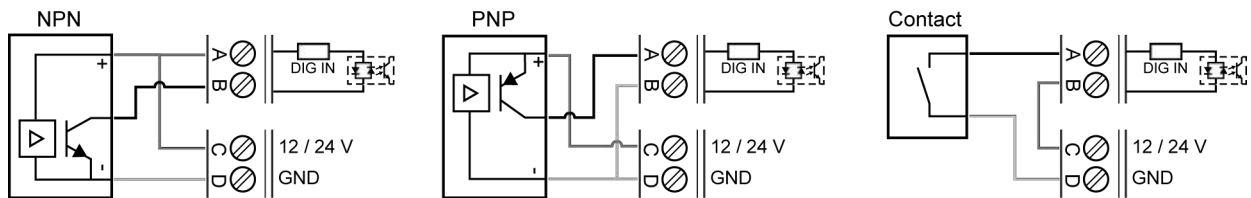


Fig. 15: CAN-IO-7-7 - Digital inputs - NPN/PNP contacts

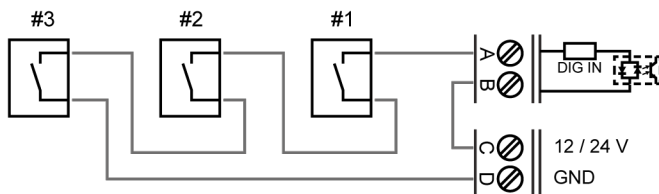


Fig. 16: CAN-IO-7-7 - Digital inputs - Thermostat contacts

Analog Outputs

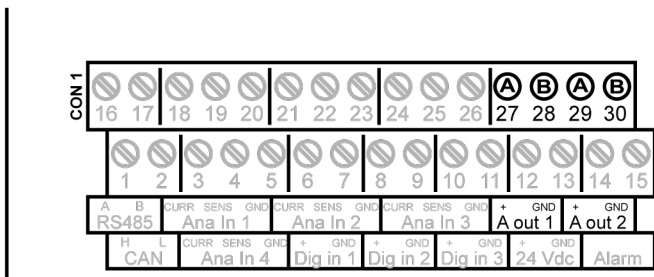


Fig. 17: CAN-IO-7-7 - Analog outputs - General

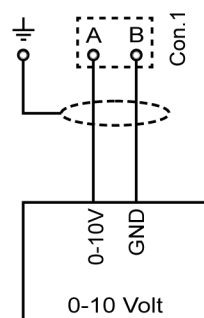


Fig. 18: CAN-IO-7-7 - Analog outputs - 0-10V

Digital Outputs

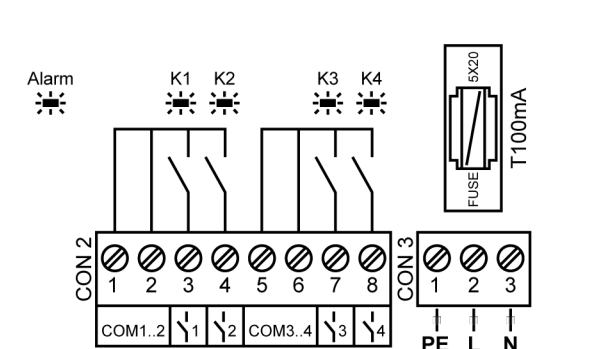
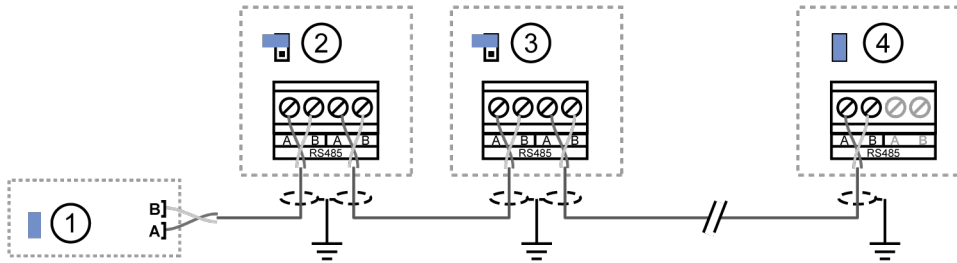


Fig. 19: CAN-IO-7-7 - Digital outputs

RS485



- ① Module 1 - RS485 bus termination
- ② Module 2 - RS485 bus termination
- ③ Module 3 - RS485 bus termination
- ④ Last module - RS485 bus termination



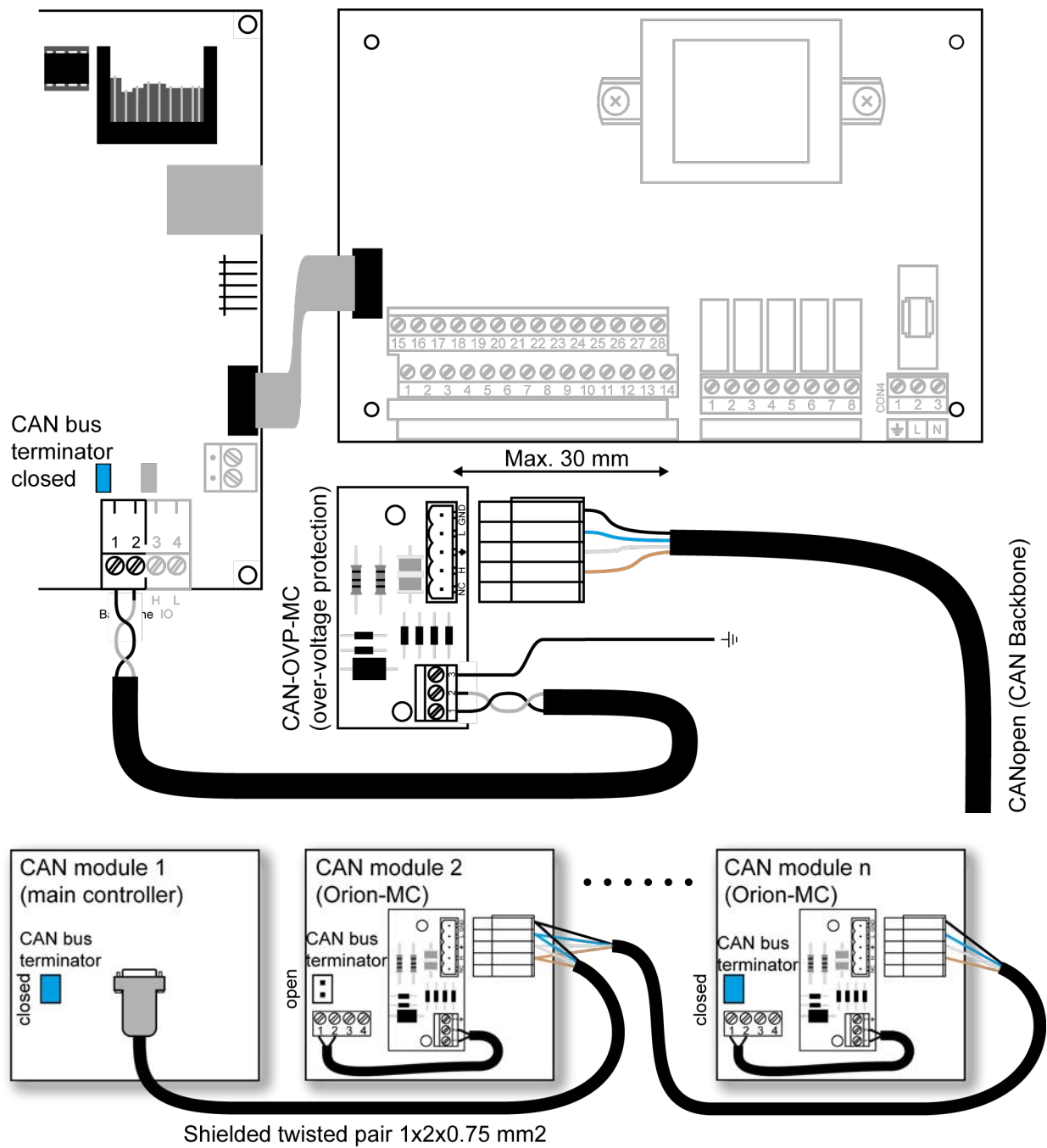
Terminate the RS485 connection with jumpers at the first and last communication node.

4.4 Make the CANopen Connection

You can use CANopen for the communication between the main controller (climate computer) and the Orion-MC. CANopen communication occurs through the CAN Backbone data bus.

🔗 „CANopen“ on page 7

You can connect two or more Orion-MC controllers to the main controller.



4.5 Make the RS232 Connection (PC and/or Modem)

Personnel: ■ Installer



Put the jumper in position **COM1** to activate the RS232 port (USB is deactivated).

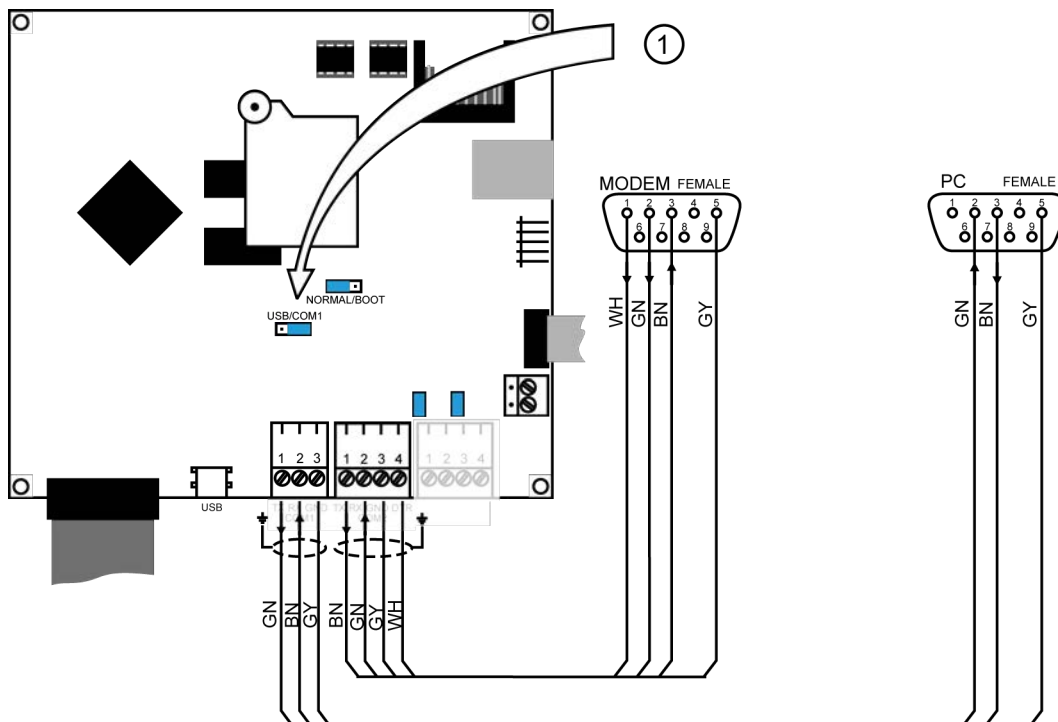


Fig. 20: RS232 - Terminal connection

① Jumper to COM1

4.6 Make the USB Connection (PC)

Personnel: ■ Installer



Put the jumper in position **USB** to activate the USB port (COM1 is deactivated).

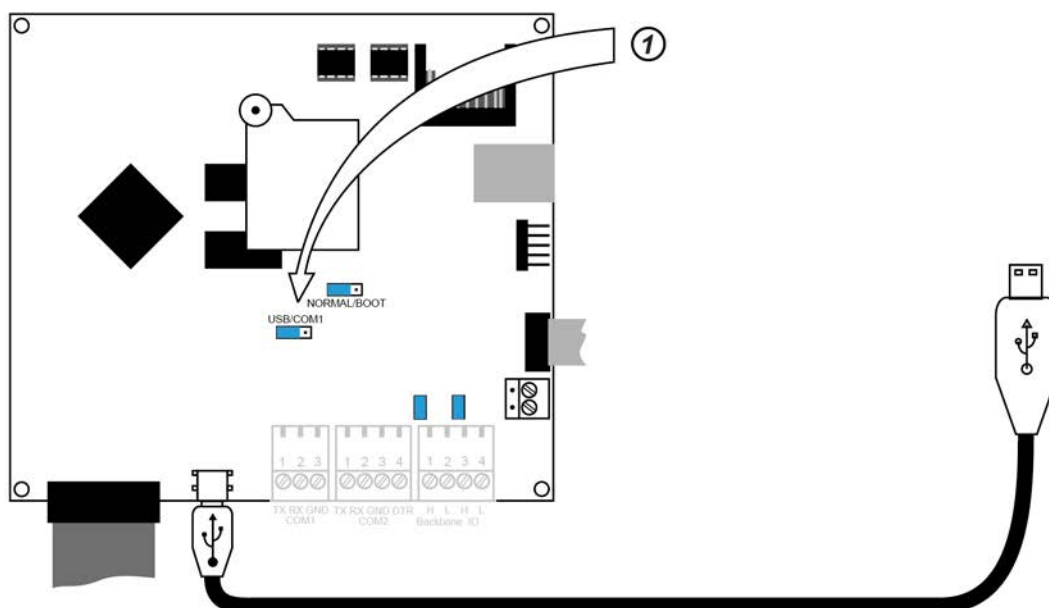


Fig. 21: USB - Connection

- ① Set the jumper to USB

4.7 Make the Ethernet Connection (TCP/IP)

Personnel: ■ Installer



Required cable to the router: Cat5e shielded Ethernet cable (max. 100 m).

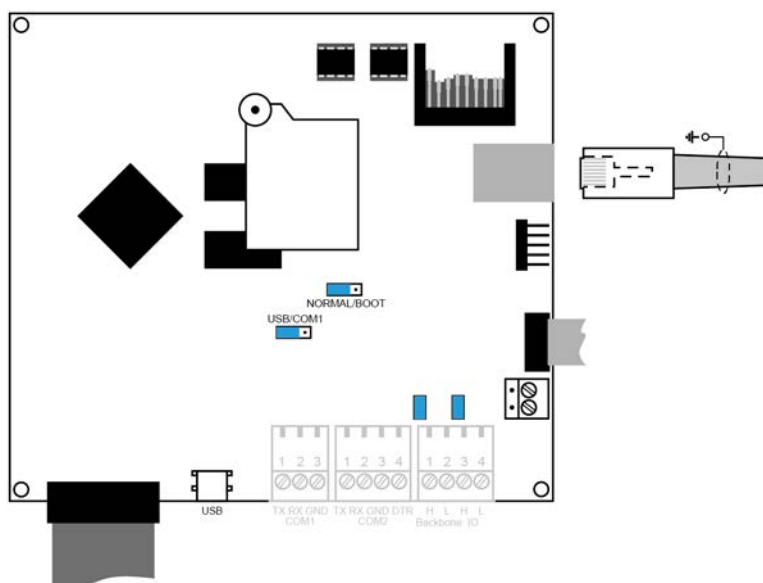


Fig. 22: Ethernet - RJ45 connection

4.8 Insert and Remove the SD Card

Personnel: Installer



Put the SD card into the SD card slot on the CPU board. The Orion-MC supports SD cards that are formatted as FAT16 or FAT32.

Before you remove an SD card from the SD card slot, make sure that you unmount the card through the software:

- Press .
- Press  to select „SD CARD“.
- Press  two times.
- Press  to select „remove“.
- Press **OK**.

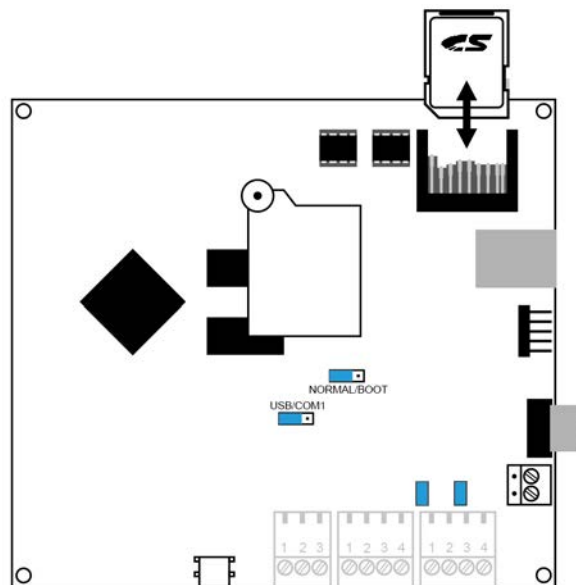


Fig. 23: Insert or remove the SD card

4.9 Make the Installation Settings (Options)

4.9.1 Installation Settings - General

Introduction

Personnel: Installer



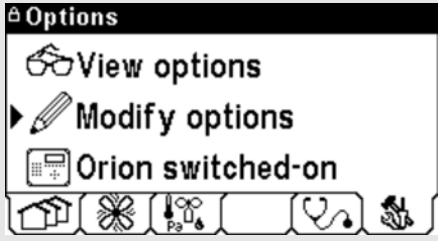
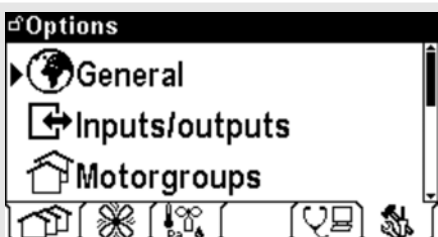
Installation level: „F6 → General“

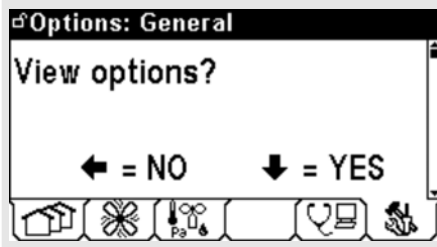
When the wiring of Orion-MC is ready, you must set the desired controls.



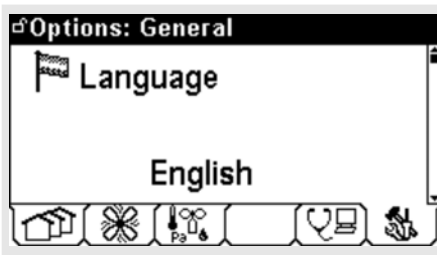
The screenshots are included for illustrative purposes. The exact settings and values on your Orion-MC can be different.

Settings

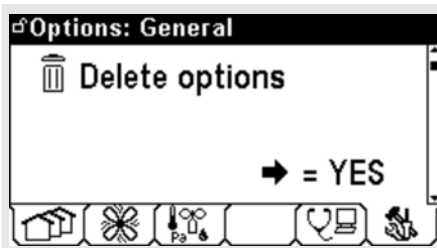
Screen	Description
	• Press F6. • Select „Modify options“ with . • Press .
	Go to the settings for „General“ with .



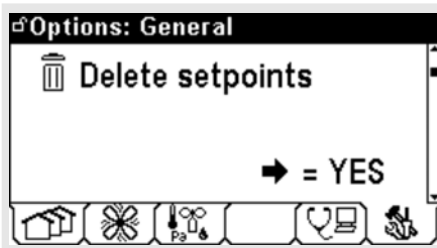
- ←: Go back to the previous screen.
- ↓: Change the settings.



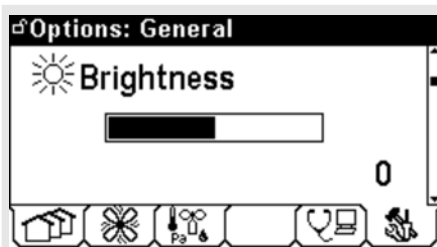
Select the desired language for the display.



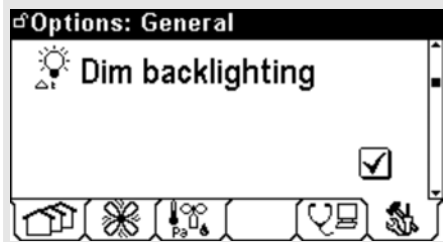
Delete the existing options in the Orion-MC.



Delete the existing settings in the Orion-MC.



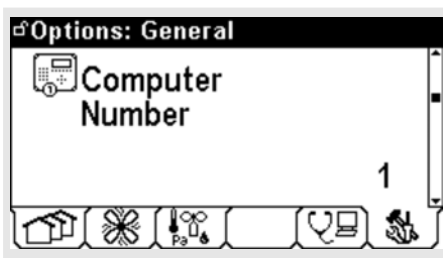
Set the brightness of the display.



This function decreases the backlight of the display automatically when the Orion-MC was not operated for one minute.

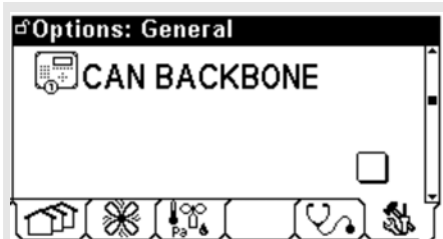
Enable/disable the function with ▲:

- ☒ = ON
- ☐ = OFF



Enter the address of the computer on the CAN backbone (computer number). It is recommended to set a higher number, so you can use the lower number for the rooms.

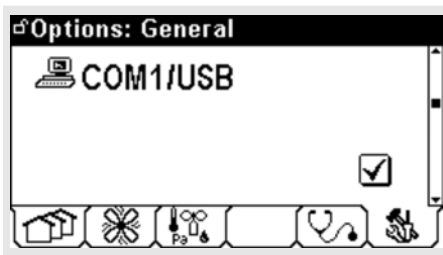
Example: The Orion-MC has number 101. The rooms have the numbers 1 to 8.



When you use CANopen, you must enable „CAN BACKBONE“. Then the controllers on the backbone can send and/or receive data, for example data from the weather station. The CAN backbone has a fixed speed of 50 kBd.

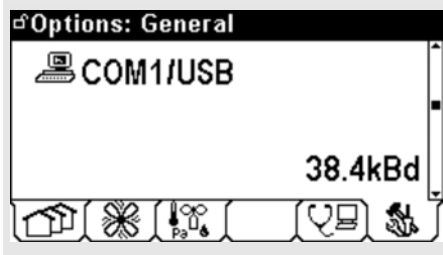
Enable/disable the function with ▲:

- ☒ = ON
- ☐ = OFF

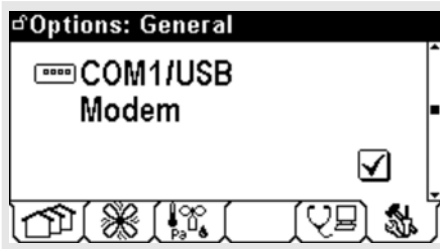


Enable/disable the function with ▲:

- ☒ = ON
- ☐ = OFF

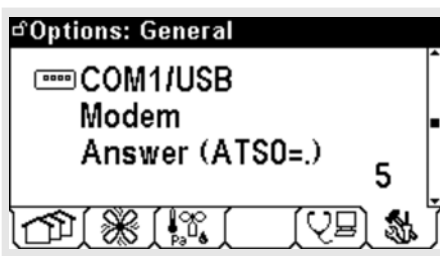


Select the communication speed with ▲.

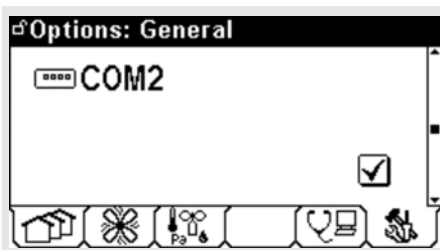


Enable/disable the function with ▲:

- ☒ = ON
- ☐ = OFF

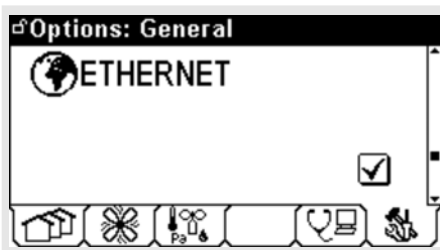


When a modem is connected, set the number of rings before the modem answers with the numeric keys or ▲▼.



Enable/disable the function with ▲:

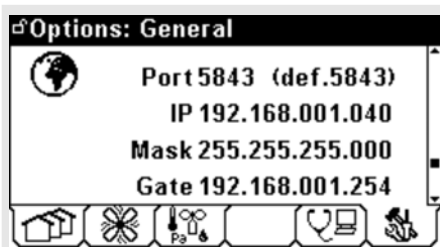
- ☒ = ON
- ☐ = OFF



When „Ethernet“ is enabled, the Orion-MC uses the TCP/IP protocol.

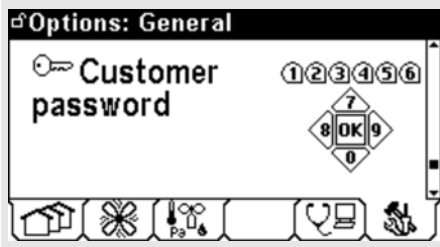
Enable/disable the function with ▲:

- ☒ = ON
- ☐ = OFF



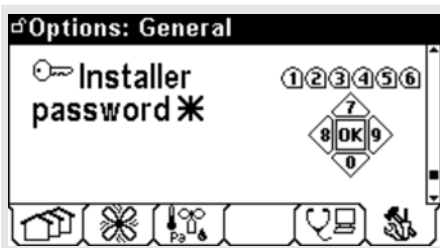
Set the parameters with ▲▼.

Press ► to navigate through the other parameters.



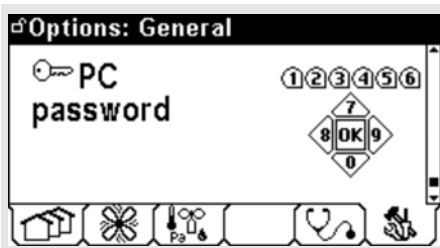
When an installer password is set, it is possible to set the rights for maximum four users.

Enter a code for each user with the keys displayed. A password must have 1-4 digits. Do not start with 0 because 0 disables the password function.

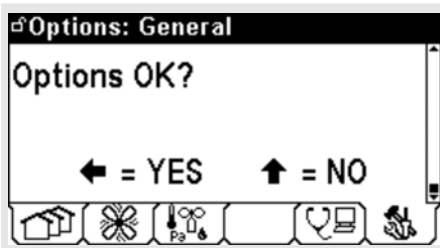


The installer password makes sure that only authorized persons can change the options under **F6**. Other persons can only view the options under **F6**.

Enter an access code for the installer or administrator with the keys displayed. A password must have 1-4 digits. Do not start with 0 because 0 disables the password function.



Enter an access code for users that log in from PC software, Rainbow+ for example.



- ←: Finish the settings and go back to the previous menu item.
- ↑: Go back to changing the current settings.

4.9.2 Installation Settings - Inputs/Outputs

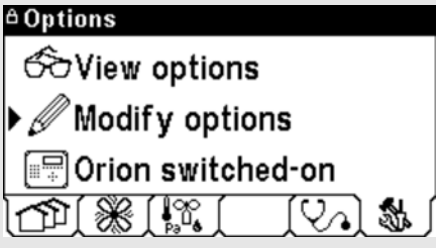
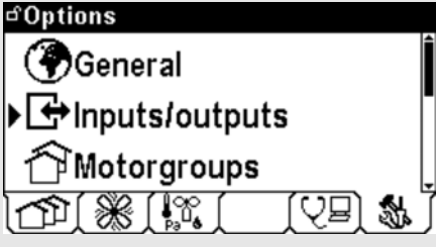
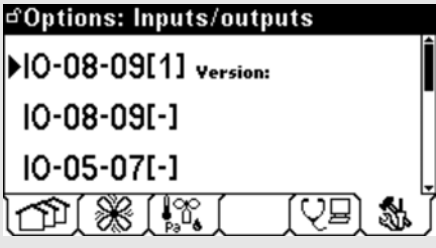
4.9.2.1 Installation Settings - Inputs/Outputs CAN-IO-8-9

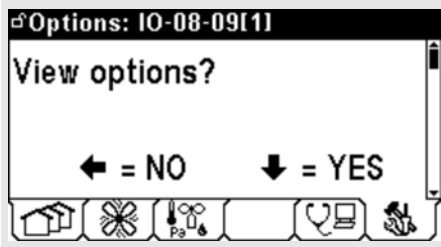
Personnel: ☒ Installer



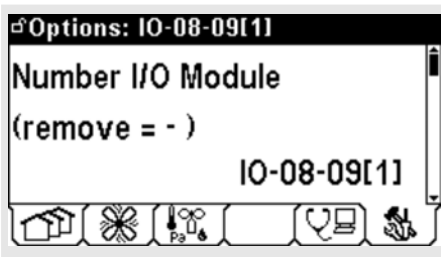
Installation level: „F6 → Inputs/Outputs“

Settings

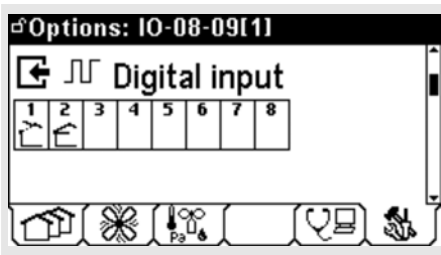
Screen	Description
	- Press F6. - Select „Modify options“ with . - Press .
	- Select „Inputs/outputs“ with . - Go to the I/O modules with .
	- Select the I/O module for which you want to change the settings with . - Go to the settings for the I/O module with .



- ←: Go back to the previous screen.
- ↓: Change the settings.



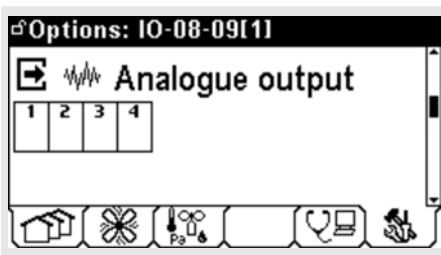
Set the address of the I/O module with ▲ ▼.



Select the function of each digital input with ▲.
To finish the settings, press ► several times until no number flashes.



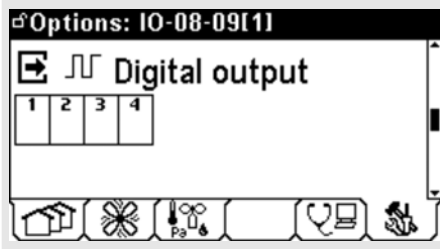
The Orion-MC shows the status of the input („ON“ or „OFF“).



Select the function of each analog output with ▲.
To finish the settings, press ► several times until no number flashes.



You can configure the analog outputs. First select the function, then press **OK**. Next, select „0-5V“ or „0-20 mA“ with ▲ ▼. After that, press **OK** to set the minimum and maximum of the output.

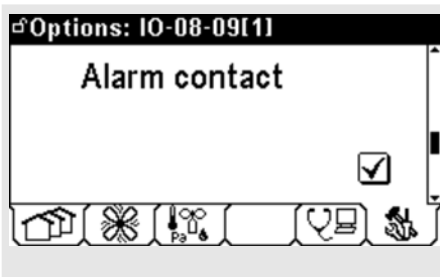


Select the function of each digital output with ▲.

To finish the settings, press ► several times until no number flashes.



Do a check on the output with ▲ ▼.

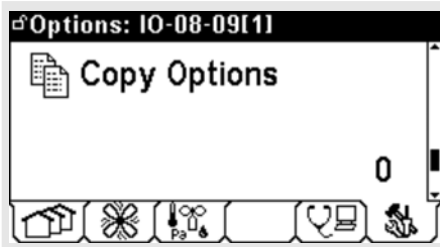


When the alarm contact is enabled, it opens when an alarm occurs. Orion-MC gives an alarm message.

When the alarm contact is disabled, it opens only when the CAN-Local communication is broken, or when a power failure occurs.

Enable/disable the function with ▲:

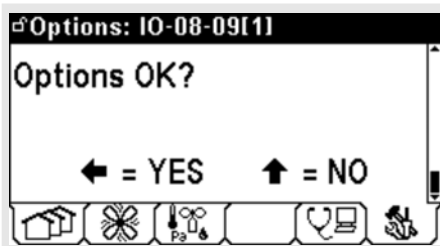
- ☒ = ON
- ☐ = OFF



You can copy the options under F6 from this I/O module to another.

Select the CAN-Local address of the target I/O module with ▲.

The new I/O module gets the same options. You can continue with the options of the new I/O module.



- ←: Finish the settings and go back to the previous menu item.
- ↑: Go back to changing the current settings.

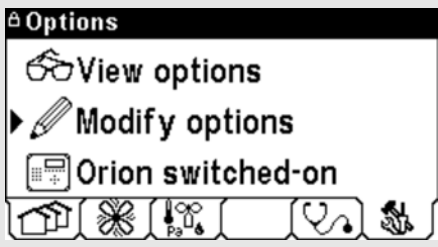
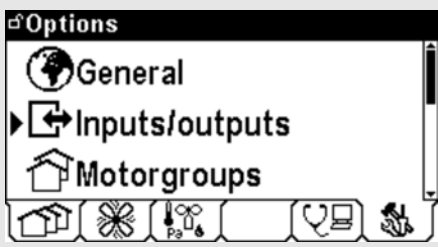
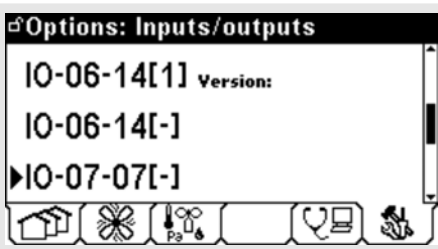
4.9.2.2 Installation Settings - Inputs/Outputs CAN-IO-7-7

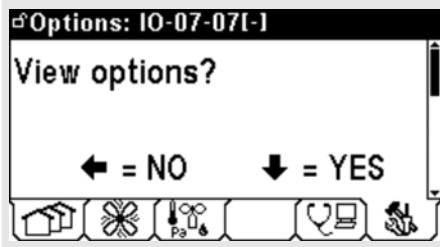
Personnel: ■ Installer



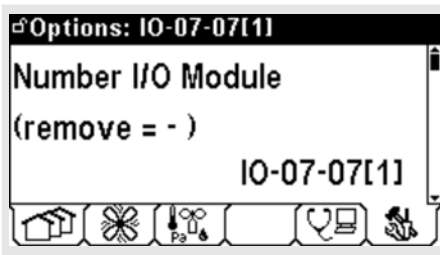
Installation level: „F6 → Inputs/Outputs“

Settings

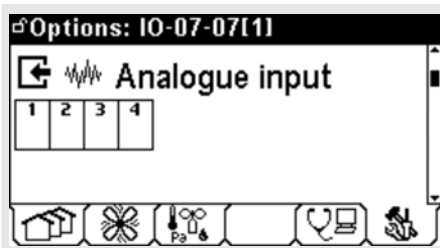
Screen	Description
	- Press F6. - Select „Modify options“ with . - Press .
	- Select „Inputs/outputs“ with . - Go to the I/O modules with .
	- Select the I/O module for which you want to change the settings with . - Go to the settings for the I/O module with .



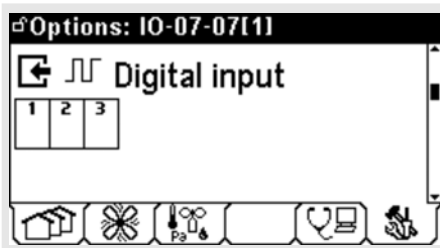
- ←: Go back to the previous screen.
- ↓: Change the settings.



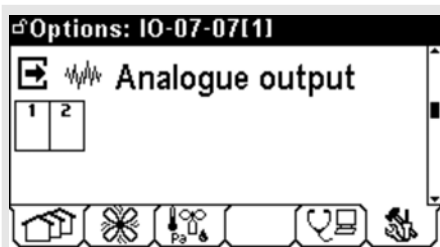
Set the address of the I/O module with ▲ ▼.



Select the function of each analog input with ▲.
To finish the settings, press ► several times until no number flashes.



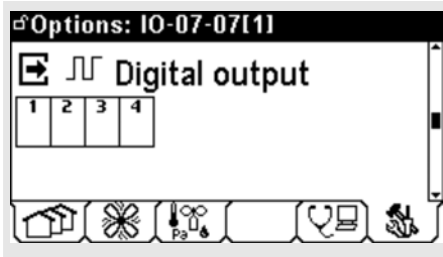
Select the function of each digital input with ▲.
To finish the settings, press ► several times until no number flashes.



Select the function of each analog output with ▲.
To finish the settings, press ► several times until no number flashes.



Press **OK** to set the minimum and maximum of the output (voltage).

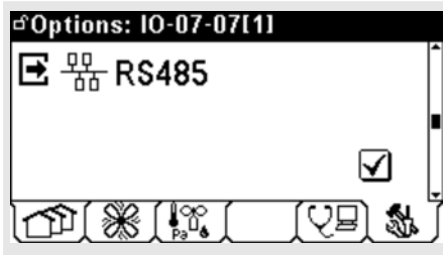


Select the function of each digital output with ▲.

To finish the settings, press ► several times until no number flashes.



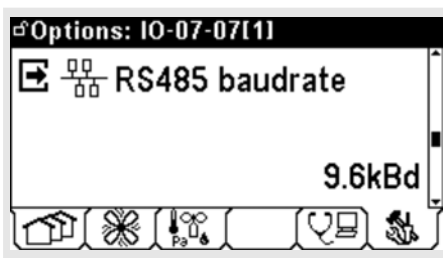
You can set each output ON or OFF, and you can test the outputs.



When the RS485 connection is enabled, you can use this connection for the communication with the fans, for example.

Enable/disable the function with ▲:

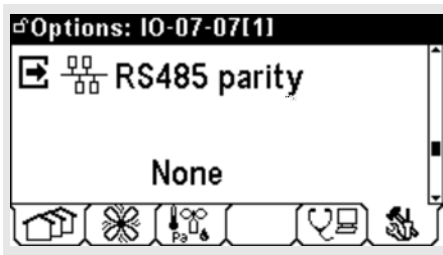
- ☒ = ON
- ☐ = OFF



Only when „RS485“ is enabled.

Select the speed of the RS485 connection with ▲:

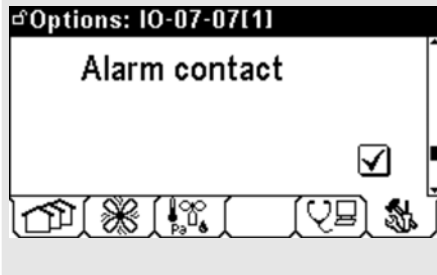
- ebmBus: 9.6 kBd
- Modbus Ziehl-Abegg: 19.2 kBd
- Modbus EBM: 19.2 kBd



Only when „RS485“ is enabled.

Select the parity of the RS485 connection with ▲.

- ebmBus: „None“
- Modbus Ziehl-Abegg: „Even“
- Modbus EBM: „Even“

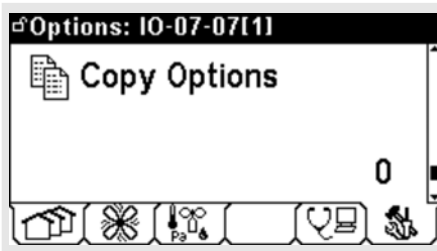


When the alarm contact is enabled, it opens when an alarm occurs. Orion-MC gives an alarm message.

When the alarm contact is disabled, it opens only when the CAN-Local communication is broken, or when a power failure occurs.

Enable/disable the function with :

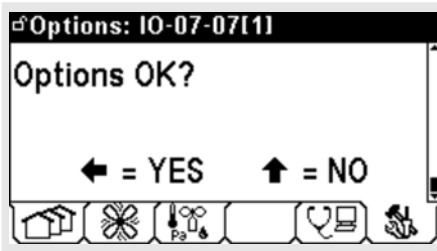
- ☒ = ON
- ☐ = OFF



You can copy the options under  from this I/O module to another.

Select the CAN-Local address of the target I/O board with .

The new I/O module gets the same options. You can continue with the options of the new I/O module.



- ←: Finish the settings and go back to the previous menu item.
- ↑: Go back to changing the current settings.

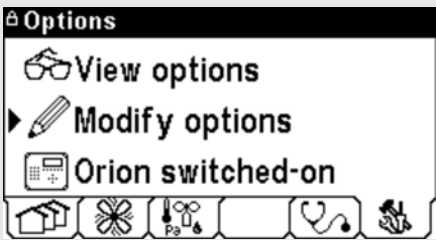
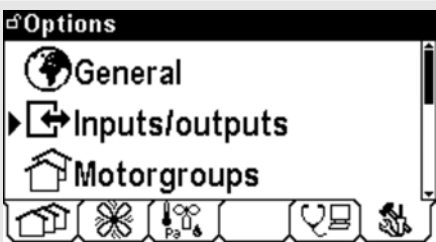
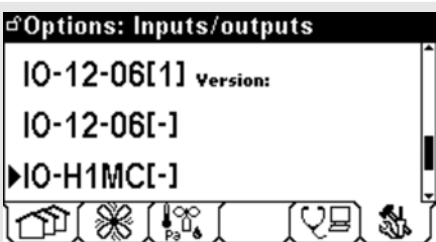
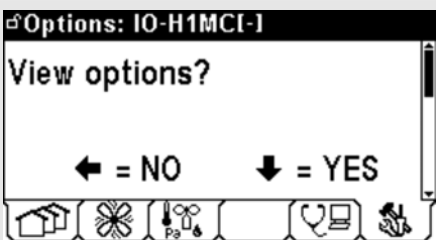
4.9.2.3 Installation Settings - Inputs/Outputs IO-H1MC

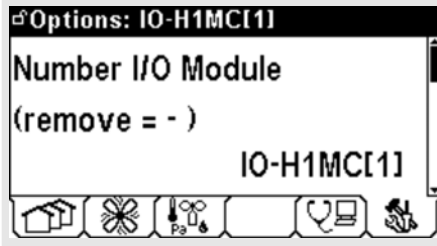
Personnel:  Installer



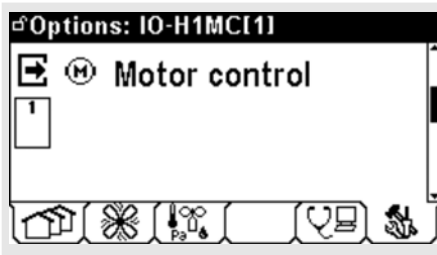
Installation level: „F6 → Inputs/Outputs“

Settings

Screen	Description
	• Press F6. • Select „<i>Modify options</i>“ with . • Press .
	• Select „<i>Inputs/outputs</i>“ with . • Go to the I/O modules with .
	• Select the I/O module for which you want to change the settings with . • Go to the settings for the I/O module with .
 The options for the H2MC are the same.	
	• : Go back to the previous screen. • : Change the settings.



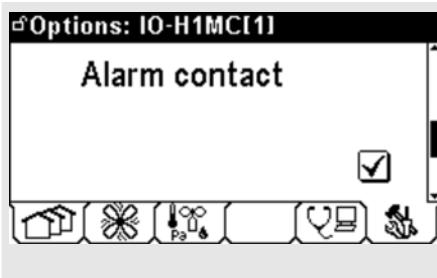
Set the CAN-Local address of the I/O module with ▲ ▼.



Select the correct type of motor:

- Window motor (100% = open, 0% = closed)
- Screen motor (100% = open, 0% = closed)

You can also see the current setting for the position here.

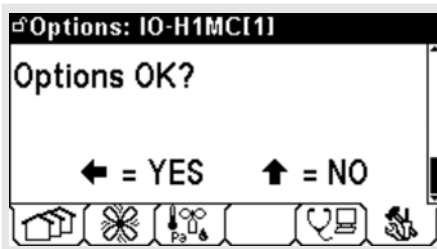


When the alarm contact is enabled, it opens when an alarm occurs. Orion-MC gives an alarm message.

When the alarm contact is disabled, it opens only when the CAN-Local communication is broken, or when a power failure occurs.

Enable/disable the function with ▲ :

- ☒ = ON
- ☐ = OFF



- ←: Finish the settings and go back to the previous menu item.
- ↑: Go back to changing the current settings.

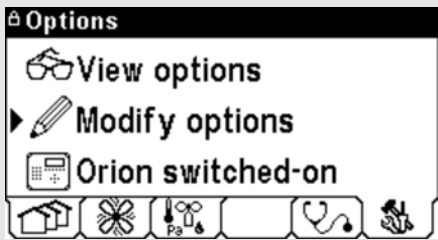
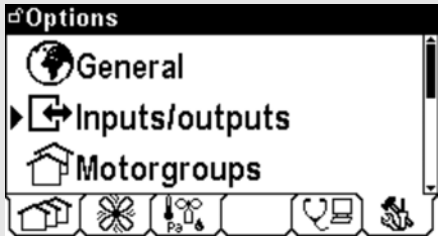
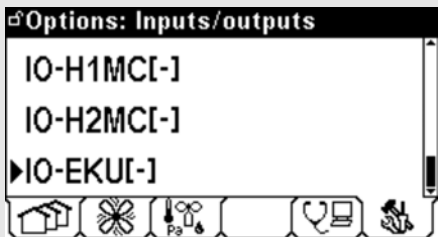
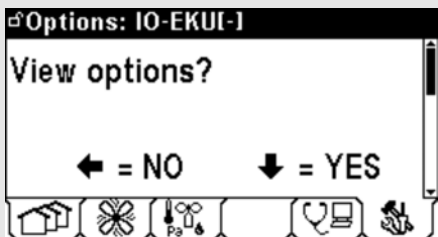
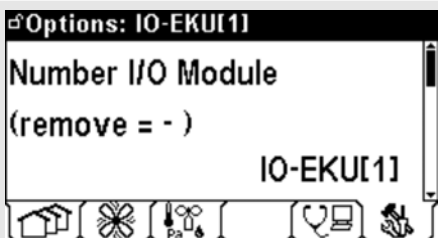
4.9.2.4 Installation Settings - Inputs/Outputs IO-EKU

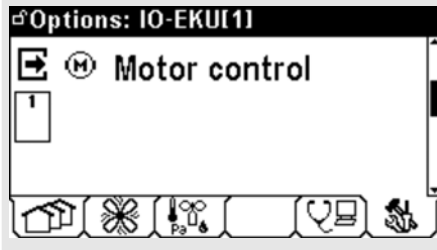
Personnel: ■ Installer



Installation level: „F6 → Inputs/Outputs“

Settings

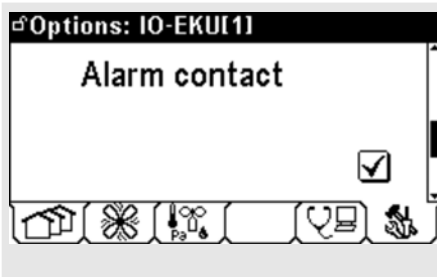
Screen	Description
	- Press F6. - Select „<i>Modify options</i>“ with . - Press .
	- Select „<i>Inputs/outputs</i>“ with . - Go to the I/O modules with .
	- Select the I/O module for which you want to change the settings with . - Go to the settings for the I/O module with .
	 = NO  = YES : Go back to the previous screen. : Change the settings.
	Set the address of the I/O module with   .



Select the correct type of motor:

- Window motor (100% = open, 0% = closed)
- Screen motor (100% = open, 0% = closed)

You can also see the current setting for the position here.

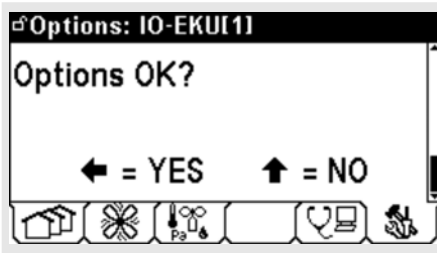


When the alarm contact is enabled, it opens when an alarm occurs. Orion-MC gives an alarm message.

When the alarm contact is disabled, it opens only when the CAN-Local communication is broken, or when a power failure occurs.

Enable/disable the function with ▲:

- ☒ = ON
- ☐ = OFF



- ←: Finish the settings and go back to the previous menu item.
- ↑: Go back to changing the current settings.

4.9.3 Installation Settings - Motor Groups

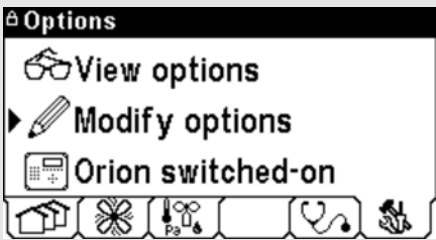
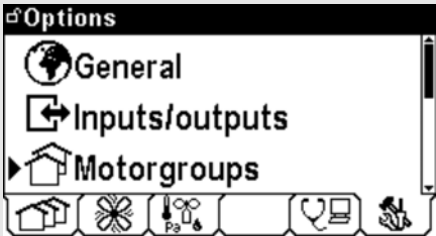
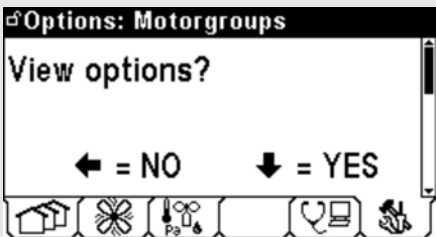
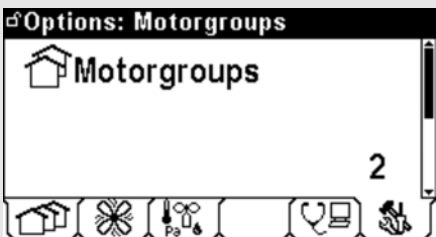
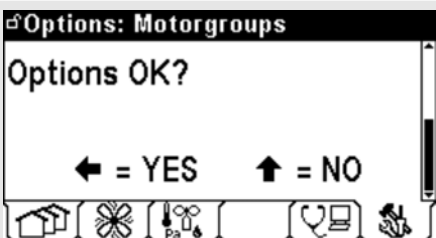
Introduction

Personnel: ■ Installer



Installation level: „F6 → *Motorgroups*“

Settings

Screen	Description
	- Press F6. - Select „Modify options“ with . - Press .
	- Select „Motorgroups“ with . - Go to the I/O modules with .
	: Go back to the previous screen. : Change the settings.
	Select the number of motor groups with  (maximum 32).
	: Finish the settings and go back to the previous menu item. : Go back to changing the current settings.

4.9.4 Installation Settings - Group Settings (Flaps)

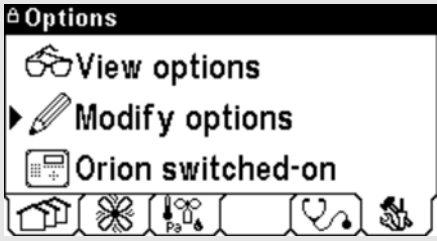
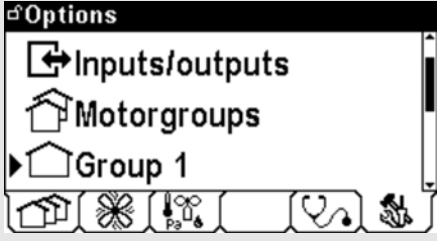
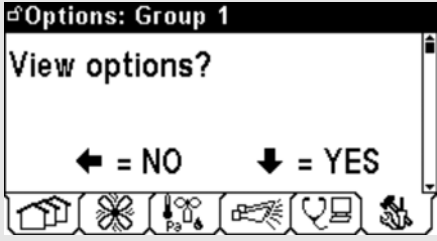
Introduction

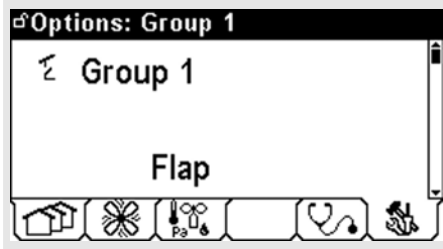
Personnel: Installer



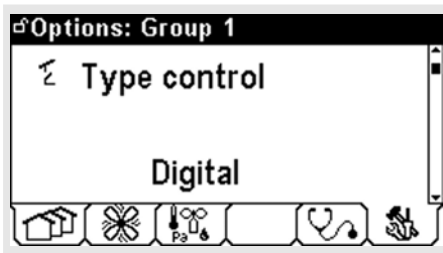
Installation level: „F6 → Group 1“

Settings

Screen	Description
	• Press F6. • Select „Modify options“ with . • Press .
	• Select „Group 1“ with . • Go to the settings with .
	• ←: Go back to the previous screen. • ↓: Change the settings.

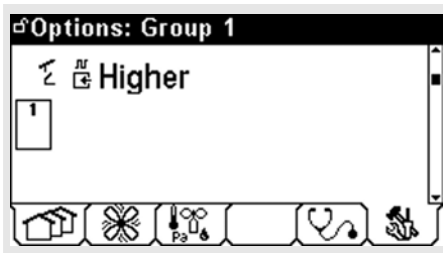


Select the component that this group controls: „*Flap*“.



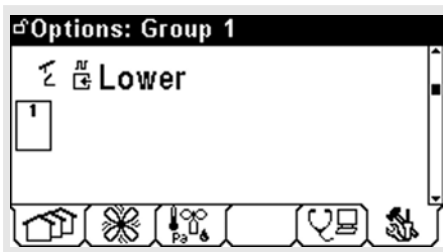
Select the type of signals from the main controller to the Orion-MC:

- „*Digital*“
- „*Analog*“



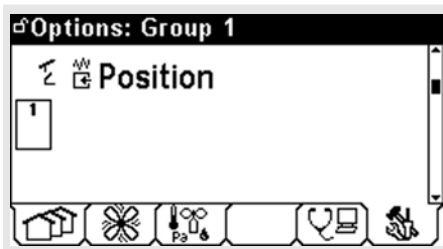
Only when the type of control is set to „Digital“.

Assign the digital input for a higher speed to the correct I/O module.



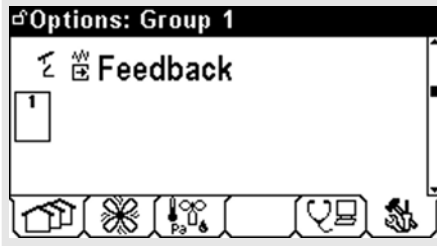
Only when the type of control is set to „Digital“.

Assign the digital input for a lower speed to the correct I/O module.

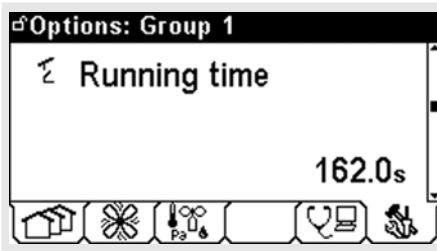


Only when the type of control is set to „Analog“.

Assign the analog input for the position to the correct I/O module.

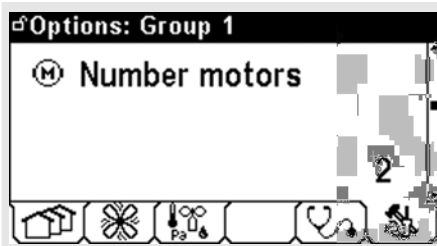


Assign the analog output when feedback on the position of the flaps to the main controller is necessary.

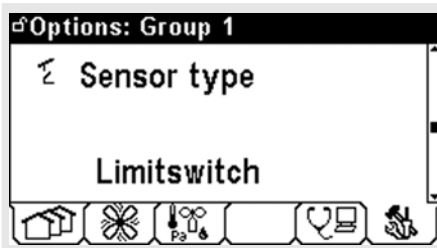


Only when the type of control is set to „Digital“.

Set the run time that is necessary to take the flap motors from 0% to 100%. The run time is important for the feedback to the main controller. A longer run time increases the accuracy of the feedback.

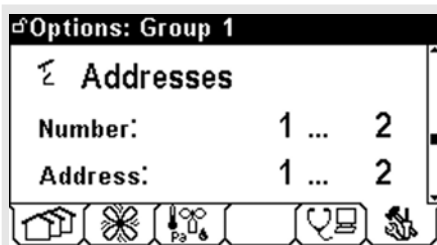


Select the number of flap motors in this group.



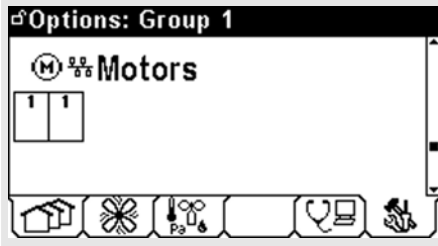
Select the type of sensor:

- „None“
- „Limit switch“
- „External alarm“



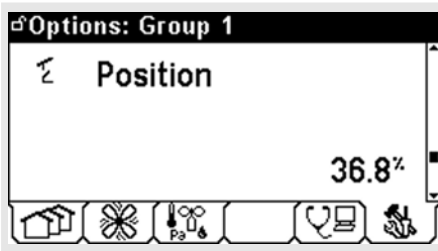
Set the numbers of the flap motors in this group and their addresses.

The flap motors are numbered sequentially. When group 1 has 20 flap motors, these are numbered 1 ... 20. When group 2 has 14 flap motors, these are numbered 21-24, and so on.



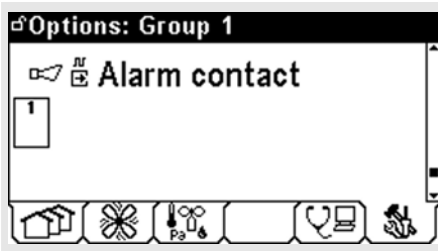
Select/cancel the I/O module to which the fans in this group are connected with ▲▼:

- ☒ = selected
- ☐ = cancelled



Only when the type of control is set to „Digital“.

The current flap position in % open.



When the alarm contact is enabled, it opens when an alarm occurs. Orion-MC gives an alarm message.

Enable/disable the function with ▲:

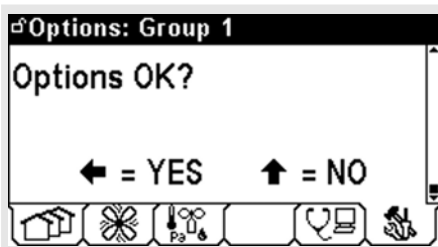
- ☒ = ON
- ☐ = OFF



Only when „Sensor type“ is set to „Limit switch“ or „External alarm“.

Select/cancel the I/O module to which the alarm is connected with ▲▼:

- ☒ = selected
- ☐ = cancelled



- ←: Finish the settings and go back to the previous menu item.
- →: Go back to changing the current settings.

4.9.5 Installation Settings - Group Settings (Ventilation)

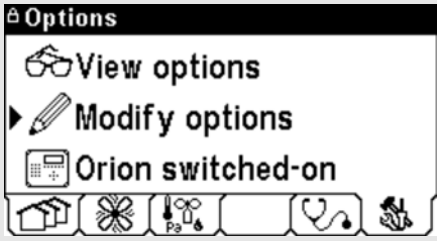
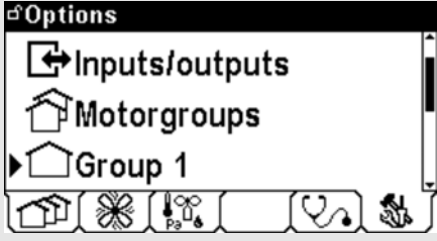
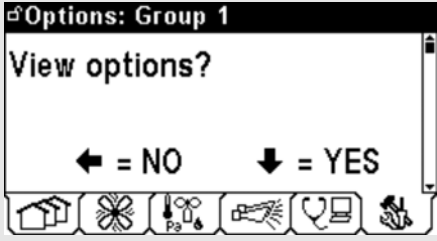
Introduction

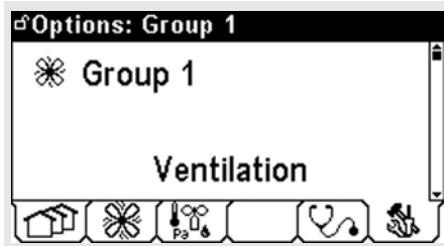
Personnel: ■ Installer



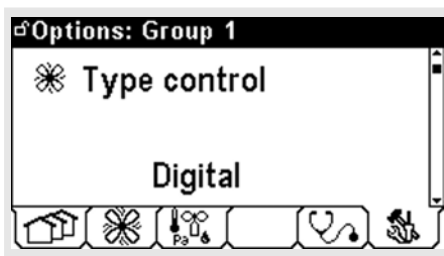
Installation level: „F6 → Group 1“

Settings

Screen	Description
	- Press F6. - Select „Modify options“ with . - Press .
	- Select „Group 1“ with . - Go to the settings with .
	←: Go back to the previous screen. ↓: Change the settings.

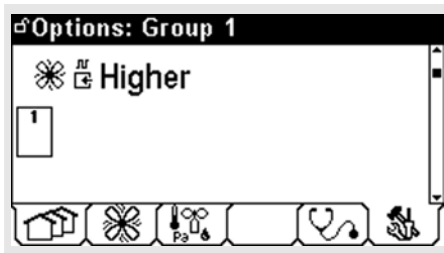


Select the component that this group controls: „Ventilation“.



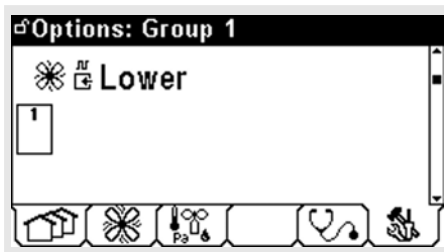
Select the type of signals from the main controller to the Orion-MC:

- „Digital“
- „Analog“



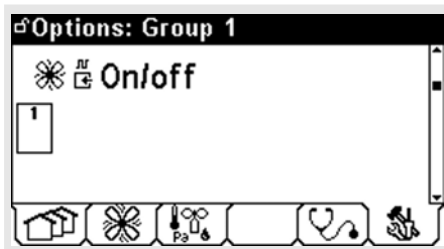
Only when the type of control is set to „Digital“.

Assign the digital input for a higher speed to the correct I/O module.



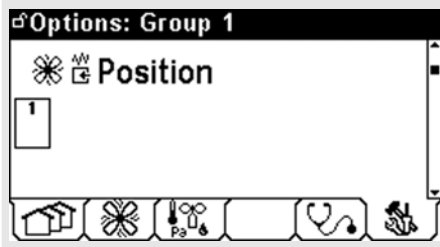
Only when the type of control is set to „Digital“.

Assign the digital input for a lower speed to the correct I/O module.



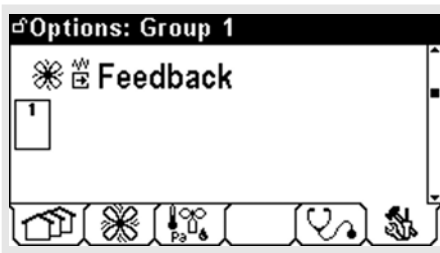
Only when the type of control is set to „Digital“.

Assign the digital input for the ON / OFF function to the correct I/O module.



Only when the type of control is set to „Analog“.

Assign the analog input for the position to the correct I/O module.

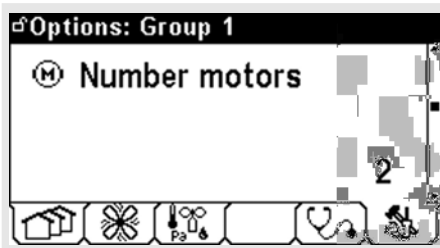


Assign the analog output when feedback on the position of the ventilation to the main controller is necessary.

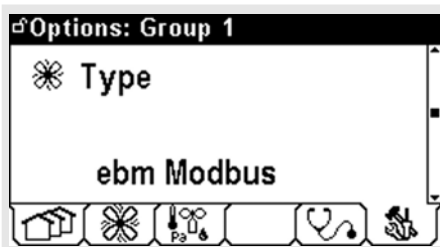


Only when the type of control is set to „Digital“.

Set the run time that is necessary to take the fans from 0% to 100%. The run time is important for the feedback to the main controller. A longer run time increases the accuracy of the feedback.

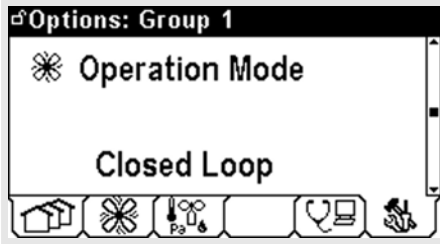


Select the number of fans in this group.



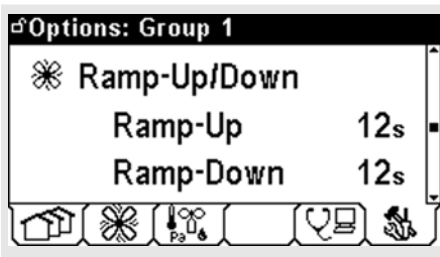
Select the communication protocol for this group:

- „ebm Modbus“
- „ebmBus“
- „Rosenberg“
- „ECblue Premium“
- „ECblue Modbus“



Select the operation mode for this group:

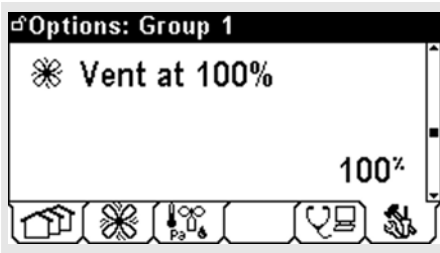
- „Open loop“ (no feedback)
- „Closed loop“ (with feedback)



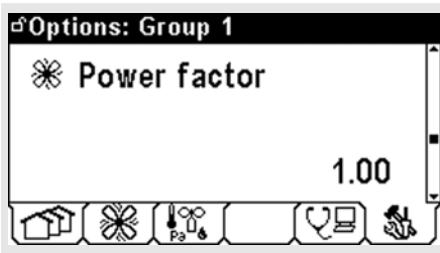
The ramp-up time is the time that is necessary for the fan to go from 0% to 100% when it is switched on.

The ramp-down time is the time that is necessary for the fan to go from 100% to 0% when it is switched off.

Select the ramp-up and ramp-down times with ▲ ▼, or the numeric keys (0 - 255 seconds).



Set the ventilation at 100% with ▲ ▼, or the numeric keys (0% ... 100%).

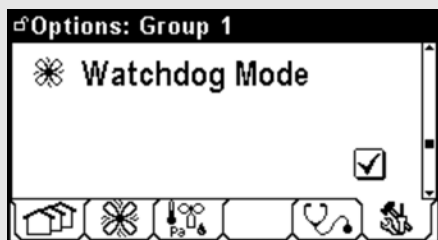


The power factor is a calculation factor to get close to the $\cos\phi$.

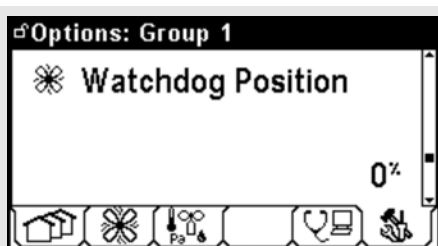
Set the power factor with ▲ ▼, or the numeric keys (0 ... 2.55).



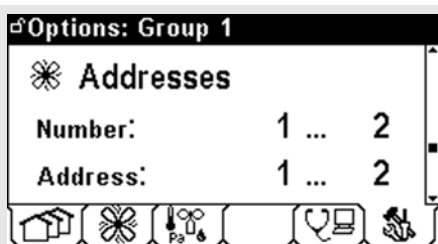
- The *Ventilation* module must be activated and selected in the installation procedure (F6) of the Orion-MC.
- The control type under „F6 → Group x“ must be set to „ebm Modbus“.



The watchdog function makes sure that the fans stay in operation at a specified ventilation percentage when the RS485 connection gets lost.



Set the ventilation percentage at which the fans must operate after the RS485 connection is lost.

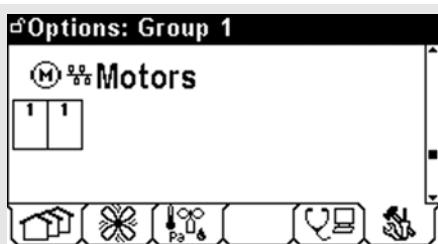


Set the numbers of the fans in this group and their addresses.

The fans are numbered sequentially. When group 1 has 20 fans, these are numbered 1 ... 20. When group 2 has 14 fans, these are numbered 21-24, and so on.

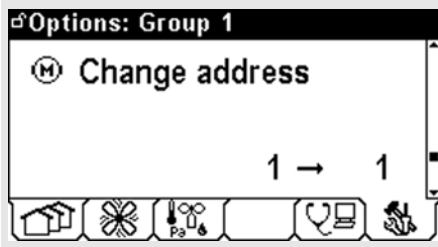


The maximum number of different RS485 buses in a group is four.



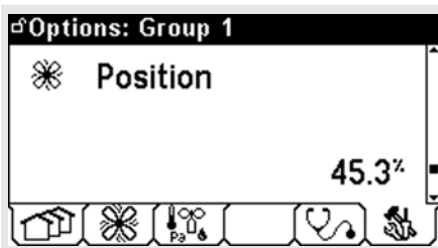
Select/cancel the I/O modules to which the fans in this group are connected with ▲ ▼:

- ☒ = selected
- ☐ = cancelled



Change the data bus address of the fans with ▲ ▼, or the numeric keys. The first number is the original number, the second number is the new number.

For the communication with the Orion-MC, you must set the address to the number of the fan +1.

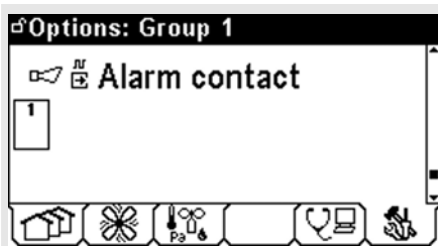


Only when „Digital“ is selected.

This setting applies when the main controller controls the position of the ventilation based on a run time. Then you must configure the output position of the main controller.

Example

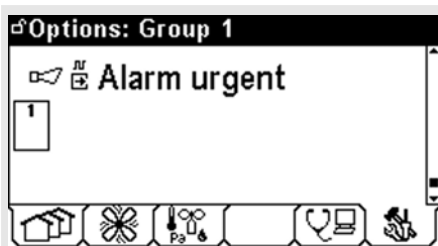
The main controller calculates that the ventilation setting must be 45.3%. Then you must enter 45.3 at „Position“.



When the alarm contact is enabled, it opens when an alarm occurs. Orion-MC gives an alarm message.

Enable/disable the function with ▲ ▼ :

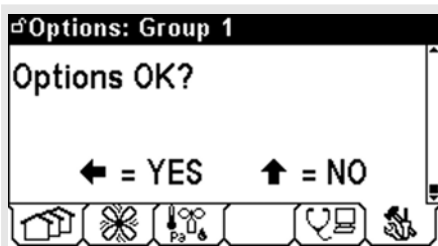
- ☒ = ON
- ☐ = OFF



When the alarm contact is enabled, it opens only when a specific number of fans report a fault. Orion-MC gives an alarm message.

Enable/disable the function with ▲ ▼ :

- ☒ = ON
- ☐ = OFF



- ←: Finish the settings and go back to the previous menu item.
- ↑: Go back to changing the current settings.

4.9.6 Installation Settings - Release Inputs Fans

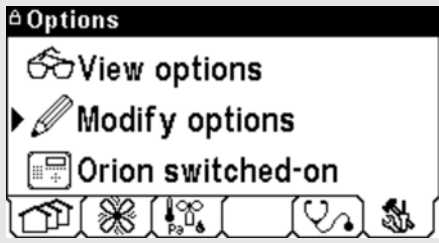
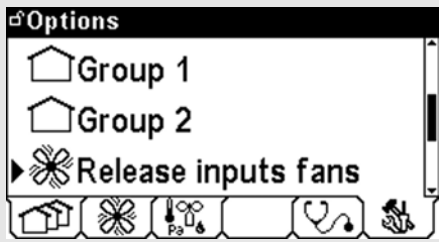
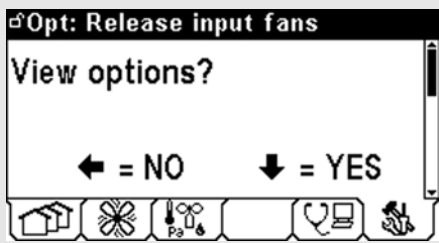
Introduction

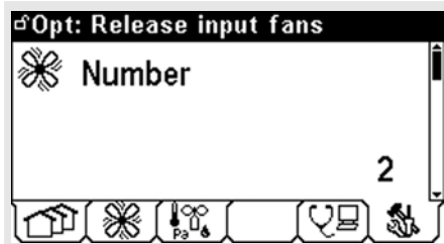
Personnel: ■ Installer



Installation level: „F6 → Release inputs fans“

Settings

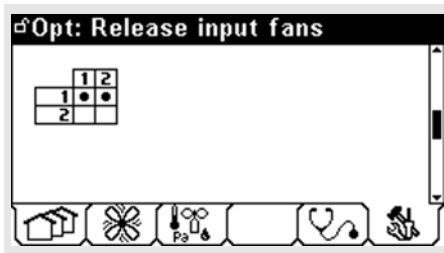
Screen	Description
	• Press F6. • Select „Modify options“ with . • Press .
	• Select „Release inputs fans“ with . • Go to the settings with .
	• ←: Go back to the previous screen. • ↓: Change the settings.



Select the number of groups (1 ... 12).



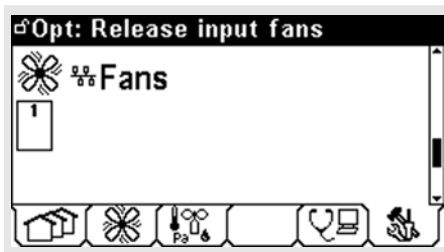
Assign the digital input for the release of the groups to the correct I/O module.



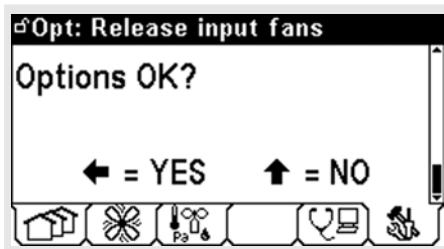
Fill out the table:

- Columns: fan numbers
- Rows: group numbers

A dot is a fan in a group. You can assign fans to one or more groups.



Assign the fans to the correct I/O module.



- ←: Finish the settings and go back to the previous menu item.
- ↑: Go back to changing the current settings.

4.9.7 Installation Settings - Alarms

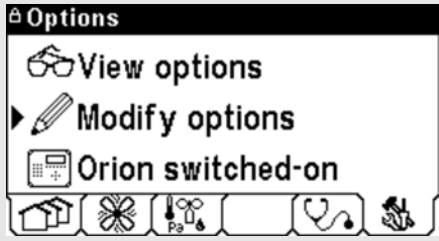
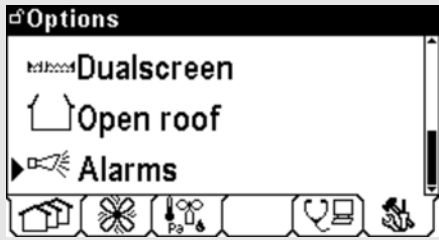
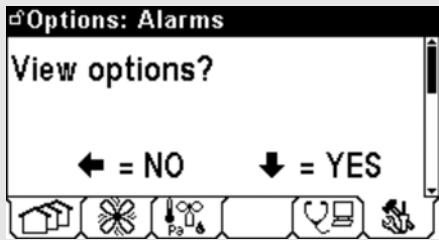
Introduction

Personnel: ■ Installer



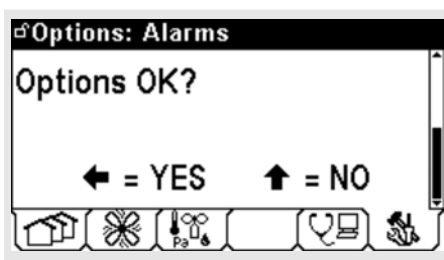
Installation level: „F6 → Alarms“

Settings

Screen	Description
	- Press . - Select „Modify options“ with . - Press .
	- Select „Alarms“ with . - Go to the settings with .
	←: Go back to the previous screen. ↓: Change the settings.



You can assign a contact to a warning here. This contact is disabled when a warning occurs.



- ←: Finish the settings and go back to the previous menu item.
- ↑: Go back to changing the current settings.

5 Operation

5.1 Basic Operation

Select and Change Controls



Personnel: ■ User

1. Press **F1**.
2. Press ▼ or ▲ to select a control.
3. At the desired control, press ►.
⇒ The settings for this control open.
4. Press ▼ or ▲ to select a setting.
5. Press ►.
⇒ The value you can change blinks and is underlined.



When you press **F6** now, you get access to the system settings. Note that not all the settings have also system settings.

6. Press ▼ or ▲ until the value is correct.
7. Confirm the value with **OK**.
8. The value stops blinking and is no longer underlined.
9. Press ▼ or ▲ to change another value, or press ◀ to go back to the main F1 menu.

Overview of the Screens

Tab. 2: Example of the screens under **F1**

F1 Main group screen	Settings	F6 System settings



The screenshots are included for illustrative purposes. The exact settings and values on your Orion-MC can be different.

Tab. 3: Description of the Screens

Screen	Description
Main group	This screen shows this information for each control: • Current values • Status of the current control You can only view the information. To change the settings, press ►
Settings	This screen shows the settings you can change for this control. • Select the setting you want to change with ▼ or ▲. • Select the value you want to change with ►. • Change the value with ▼ or ▲. • Confirm the changed value with OK. • Go back to the previous screen with ◀.
System settings	This screen shows the system settings for this control. The buttons have the same function as described above.

5.2 Operating the Orion-MC

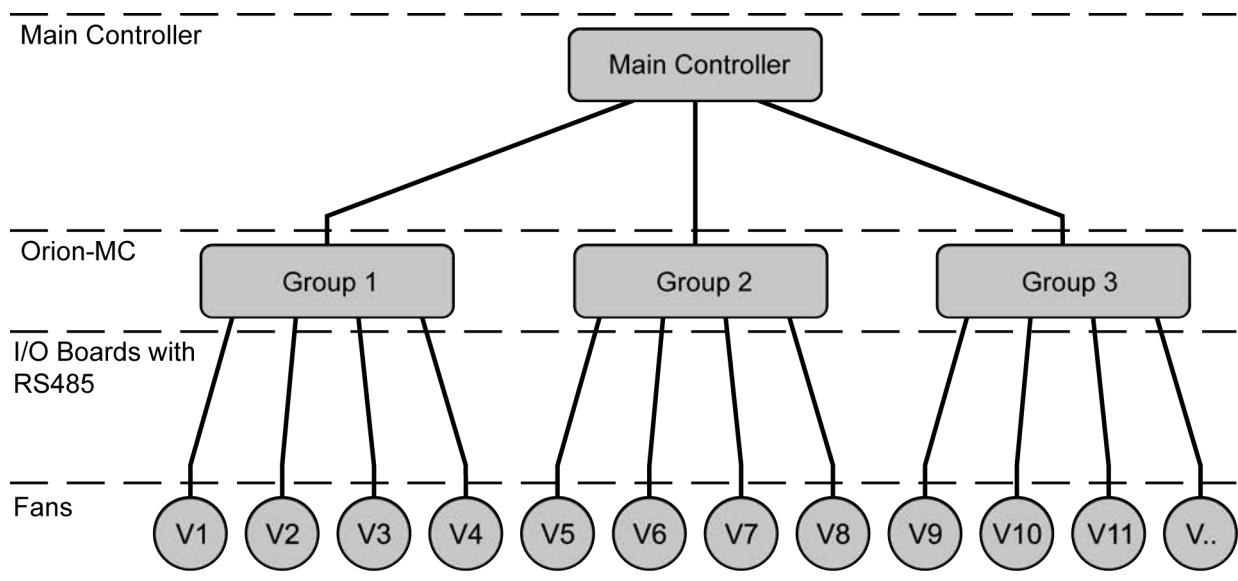
Controlling the Ventilation Systems

The greenhouse can have a number of fans (V1, V2 and so on). The fans are connected to I/O modules through RS485 communication. In turn, the I/O modules are connected to the Orion-MC.

In the Orion-MC, a number of fans are a group. Each fan group has fans that are basically controlled at the same time, and take the same position. The main controller controls the fan groups. The Orion-MC makes sure that the fans get to the correct position.

You can use the Orion-MC to control:

- all the fans in a group, or
- only a single fan, if desired.



Virtual Control

It is possible, independent of the setpoint of the motor group, to switch fans in this group on or off. For this, maximum 12 digital inputs are available. You must assign all the available fans to one or more digital inputs.

The main controller is the control unit. It determines the desired action for each group. The main controller sends the desired action to the Orion-MC. The Orion-MC simulates this action and sends feedback on the simulated action to the main controller. This is called virtual control. Then the main controller sees this feedback as a confirmation that the desired action has been carried out.

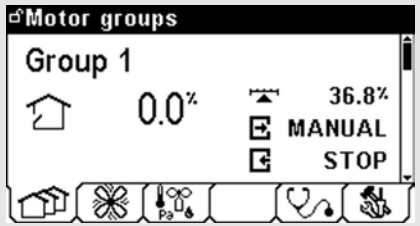
At the same time, the Orion-MC sends control commands to the I/O modules to make sure that the group carries out the desired action. The relevant units send feedback to the Orion-MC on the positions. The action stops when the correct position has been reached, or when an alarm is given because the desired position was not reached.

5.3 Flaps

5.3.1 Controls - Flaps

Introduction

Personnel: ■ User

F1 Main group	Settings	F6 System settings
 This screen shows the status of a group with flaps.	Group settings Fn.: Group 1 Operation MANUAL Position 36.8% Flap 1 0% Go to the settings for the group with . Flap motor settings Fn.: Group 1 Operation MANUAL Position 36.8% Flap 1 0% To change the settings for a single flap motor, select the flap motor with . Fn.: Flap 1 Operation AUTO Position 37% Go to the settings for the flap motor with .	Syst.: Group 1 Alarm MANUAL 30min Alarm variance 2.0% Alarm variance 5min No system settings

Main Group



Control level: „F1 → Flaps“

Status	Description
36.8%	The flap position with the biggest difference between the actual position and the setpoint.
	The desired flap position for the group that the main controller has calculated.
	The status of the group in the Orion-MC: „<i>AUTO</i>“: The group is set to automatic operation. The main controller determines the desired position. The Orion-MC makes sure that the flaps in the group go to the desired position. „<i>MANUAL</i>“: The group is set to manual operation. The Orion-MC can control the group with the „<i>Position</i>“ setting. „<i>OFF</i>“: The group is switched off. The flap position is 0%.
	The action of the main controller (for digital control): „<i>OPEN</i>“: The main controller opens the flaps of the group. „<i>CLOSE</i>“: The main controller closes the flaps of the group. „<i>STOP</i>“: The main controller stops actions.

Settings



- Control level: „F1 → Flaps → Settings“
- You can browse through the settings of the groups and flaps with .

Setting	Description
Group	
	„ <i>Operation</i> “: Change the status of the group: „<i>AUTO</i>“: The group is controlled automatically. The main controller determines the desired flap position. The Orion-MC makes sure that the flaps go to the correct position. „<i>MANUAL</i>“: The group is controlled manually. The Orion-MC can control the group with the „<i>Position</i>“ setting. „<i>OFF</i>“: The group is switched off. The flap position is 0%.
	„ <i>Position</i> “: The desired flap position for the group that the main controller has calculated. When the „ <i>Operation</i> “ setting is set to „ <i>MANUAL</i> “, you can set the flap position here.
	„ <i>Flap 1</i> “: The current position of each flap. Press  to see and change the settings of this flap.
Flap	

Setting	Description
	„Operation“: Change the status of the flap: „AUTO“: The flap motor is controlled automatically. The Orion-MC makes sure that the flap motor is in the correct position. „MANUAL“: The flap motor is controlled manually. You can set the desired flap position at the „Position“ setting. „OFF“: The flap motor is switched off. The flap position is 0%.
	„Position“: The desired position for the flap motor. When the „Operation“ setting is set to „MANUAL“, you can set the position here.

System Settings



- Control level: „F1 → Flaps → Settings (Group) → F6“
- You can browse through the settings of the groups with ◀ and ▶.
-

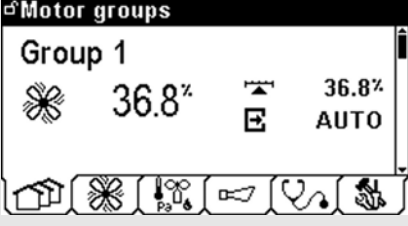
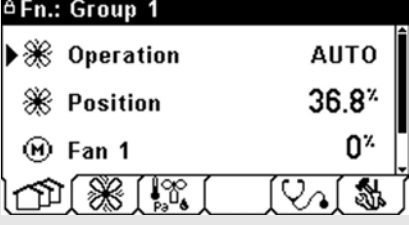
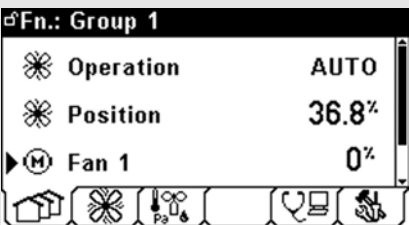
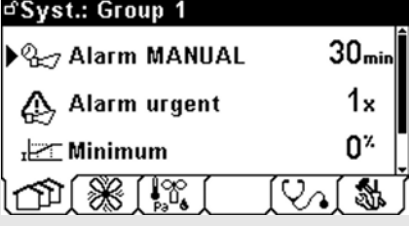
Setting	Description
	„Alarm MANUAL“: When the group on the Orion-MC is still in the „MANUAL“ mode after the set period, the alarm contact of the group is activated. When you set this value to 0, this function is OFF.
	„Alarm variance“: Here you can set the minimal variance before an alarm is given. When the difference between the actual position of the flap motor and the desired position is more than the value set here, the Orion-MC gives an alarm.
	„Alarm variance“: Here you can set the minimal wait time before an alarm is given. When the flap motor is not in the desired position after the wait time, the Orion-MC gives an alarm.

5.4 Ventilation

5.4.1 Controls - Ventilation

Introduction

Personnel: User

F1 Main group	Settings	F6 System settings
 This screen shows the status of a group with fans.	Group settings  Go to the settings for the group with . Fan settings  To change the settings for a single fan, select the fan with . Go to the settings for the fan with .	 No system settings

Main Group



Control level: „F1 → Group 1 (Ventilation)“

Status	Description
36.8%	The average ventilation position of all the fans in the group (%). This can be different from the desired position (setpoint).
	The desired ventilation position for the group that the main controller has calculated.

Status	Description
	The status of the group in the Orion-MC: • „<i>AUTO</i>“: The group is set to automatic operation. The main controller determines the desired position. The Orion-MC makes sure that the fans in the group go to the desired position. • „<i>MANUAL</i>“: The group is set to manual operation. The Orion-MC can control the group with the „<i>Position</i>“ setting. • „<i>OFF</i>“: The group is switched off. The ventilation position is 0%.

Settings



- Control level: „*F1* → *Group x (Ventilation)* → *Settings*“
- You can browse through the settings of the groups and fans with .

Setting	Description
Group	
	„<i>Operation</i>“: Change the status of the group: • „<i>AUTO</i>“: The group is controlled automatically. The main controller determines the desired ventilation position. The Orion-MC makes sure that the fans operate at the correct speed. • „<i>MANUAL</i>“: The group is controlled manually. The Orion-MC can control the group with the „<i>Position</i>“ setting. • „<i>OFF</i>“: The group is switched off. The ventilation position is 0%.
	„<i>Position</i>“: The desired ventilation position for the group that the main controller has calculated. When the „<i>Operation</i>“ setting is set to „<i>MANUAL</i>“, you can set the ventilation position here.
	„<i>Fan 1</i>“: The current ventilation position of each fan. Press  to see and change the settings of this fan.
Fan	
	„<i>Operation</i>“: Change the status of the fan: • „<i>AUTO</i>“: The fan is controlled automatically. The Orion-MC makes sure that the fan operates at the correct speed. • „<i>MANUAL</i>“: The fan is controlled manually. You can set the desired ventilation position at the „<i>Position</i>“ setting. • „<i>OFF</i>“: The fan is switched off. The ventilation position is 0%.
	„<i>Position</i>“: The desired ventilation position for the fan. When the „<i>Operation</i>“ setting is set to „<i>MANUAL</i>“, you can set the ventilation position here.

Setting	Description
	„Offset“: It is possible that a fan in a group must be set to a higher or lower ventilation position than the other fans. If necessary, you can set an incremental increase or decrease in the ventilation position of each fan (the offset). An offset is only possible when the „Operation“ setting is set to „AUTO“.

Example

Fan	Offset	Required position	Actual position
1	10%	70%	80%
2	0%	70%	70%
3	-8%	70%	62%
4	0%	70%	70%
5	8%	70%	78%

System Settings



- Control level: „F1 → Group x (Ventilation) → Settings (Group) → F6“
- You can browse through the settings of the groups with ◀ and ▶.
-

Setting	Description
	„Alarm MANUAL“: When the group on the Orion-MC is still in the „MANUAL“ mode after the set period, the alarm contact of the group is activated. When you set this value to 0, this function is OFF.
	„Alarm variance“: Here you can set the minimal variance before an alarm is given. When the difference between the actual position of the fan and the desired ventilation position is more than the value set here, the Orion-MC gives an alarm.
	„Alarm variance“: Here you can set the minimal wait time before an alarm is given. When the fan is not in the desired ventilation position after the wait time, the Orion-MC gives an alarm.
	„Alarm urgent“: You can make a distinction between the alarms for a group of fans: - an alarm from one fan (less urgent) - an alarm from a number of fans (more urgent). Two alarm contacts are assigned. The urgent alarm contact is only activated when the number of fans that generate an alarm is higher than the value you set at „Alarm urgent“. See the example below.

Setting	Description
	„<i>Minimum</i>“: With the minimum setting, you can limit the range for the ventilation positions of the fans. When the required ventilation position is lower than the minimum, then the ventilation position is the same as the minimum setting.
	„<i>Maximum</i>“: With the maximum setting, you can limit the range for the ventilation positions of the fans. When the required ventilation position is higher than the maximum, then the ventilation position is the same as the maximum setting. These limits also apply to fans that are outside these limits as a result of the offset settings for the fans.
Example	You have 140 fans. When maximum 10% of these fans fail, the process can still continue. You must set the „<i>Alarm urgent</i>“ setting to „14“. Then the urgent alarm contact is only activated when more than 14 fans give an alarm.

6 Maintenance

**WARNING!**

Electric shock can cause serious injury.

This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.

6.1 Cleaning

Personnel: ☐ Installer
☐ User

**NOTICE!**

When you clean the product incorrectly, you can damage it.

Detergents, solvents and running water can damage the product and the labels on it.

- Only use a soft cloth to clean the product.
- Only dampen the cloth with water.

6.2 Update the Software

Personnel: ☐ Installer

**WARNING!**

Electric shock can cause serious injury.

Always disconnect the power supply to the controller before you do work on the electrical components. When this is not possible, be very careful that you do not touch live parts.

**Required tools:**

- A new software file from the manufacturer.
- The FlashTool software from the manufacturer.

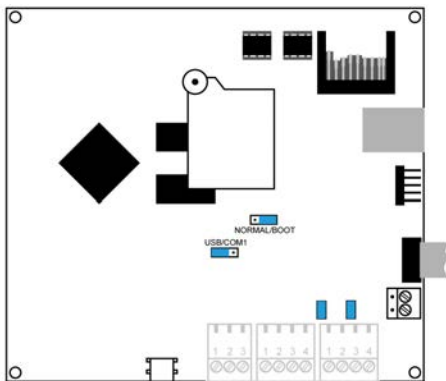


Fig. 24: Jumper - BOOT

1. ➤ Open the Orion-MC.
2. ➤ Set the jumper on the CPU board in the BOOT position.
3. ➤ Switch the Orion-MC off and on again.



Fig. 25: FlashTool

4. ➤ Start the FlashTool.
5. ➤ Examine the connection speed of the Orion-MC. Default: 115.2 kBd.
6. ➤ Select the correct COM port.
7. ➤ Enter the correct connection speed.
8. ➤ Click „CONNECT“.
9. ➤ Click „OPEN“, then browse to the hex file you want to upload.
10. ➤ Click „FLASH“.
11. ➤ When the update is ready, click „EXIT“.

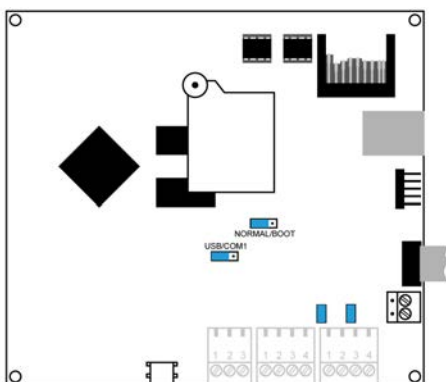


Fig. 26: Jumper - NORMAL

12. ➤ Set the jumper on the CPU board in the NORMAL position.
13. ➤ Switch the Orion-MC off and on again.

7 Troubleshooting

7.1 Alarms

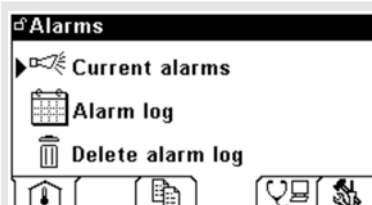
Introduction

Personnel: User

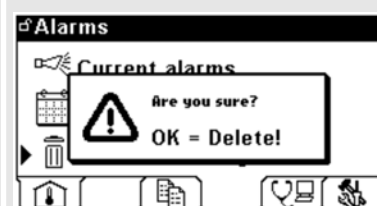
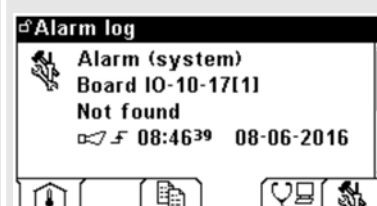


Control level: „F4 → Alarms“

Main screen



Options



Settings



Control level: „F4 → Alarms“

- „Current alarms“
See all the active alarms and warnings. You can delete or suppress some alarms or warnings.
- „Alarm log“
See the history of all the alarms and warnings.
- „Delete alarm log“
Remove the alarm history.

Types of Alarms

-  Warning: The alarm symbol flashes. The alarm contact is not broken.
-  There are two or more warnings.
-  Alarm: The alarm symbol flashes. The alarm contact is broken. This activates an alarm horn, an alarm lamp and/or a telephone dialer.
-  There are two or more alarms.
-  The alarm is suppressed.

Solve an Alarm

1. ➞ Press  for an alarm overview.
2. ➞ Scroll between the alarms with  or .
3. ➞ Press **OK** to change the type of alarm, if possible.
 -  Warning: The alarm contact is temporarily restored. When you do not solve the problem, the alarm shows again.
 -  Suppress alarm: The alarm contact is restored. The problem is not solved, but the alarm does not show again.
 -  Alarm: The alarm contact is broken.
 -  Viewed: The alarm is moved to the alarm history.
4. ➞ Find the cause of the alarm and solve the problem.
5. ➞ When the alarm is solved, press . Many alarms disappear automatically after the problem is solved.

Overview of the Orion-MC Alarms

Description	Type	Cause
Fan 'xx' Manual mode.	Alarm/Warning	Fan number 'xx' is set to manual operation mode (in the Orion-MC). The alarm contact will be disabled after a while (system setting).
Group 'xx' Manual mode.	Alarm/Warning	The group number 'xx' is set to manual operation mode in the Orion-MC. The alarm contact will be disabled after a while (system settings).
Fan 'xx' Position not reached.	Alarm	Fan number 'xx' has a different speed (system setting).

Overview of the System Alarms



Always contact your installer when a system alarm occurs!

Description	Type	Cause
Orion switched off.	Warning	The software has switched off the computer at the specified time. The controller has been switched off and gives an alarm. Press a key to suppress the alarm, and to convert the alarm into a warning. No alarms from the controllers are shown while the computer is switched off.
System restart	Warning	The computer has been restarted at the specified time.
Settings deleted.	Alarm	The settings have been deleted at the specified time (Installation procedure).
Options deleted.	Alarm	The options have been deleted at the specified time (Installation procedure).
Alarm deleted.	Warning	The alarms have been deleted at the specified time
Options restored.	Alarm	Someone has restored the options.
Settings restored.	Alarm	Someone has restored the settings.
Options + settings restored	Alarm	Someone has restored the options and settings.
Options (+ settings) not restored	Alarm	The options and settings that have been restored are not correct.

Overview of the Communication Alarms

Description	Type	Cause
Board IO-xx-xx [xx] not found.	Alarm	The Orion-MC has no connection with the relevant I/O board. Possible causes: - communication interrupted - broken contact - CAN bus terminator
No alarm contact.	Alarm	No alarm contact assigned. The IO boards are not configured correctly (Installation procedure).
Fan 'xx' not found.	Alarm	There is a communication problem with the RS485 connection (ebmBus, Modbus on the CAN-IO-7-7 board).

Overview of the Internal System Alarms

The Orion-MC does a check on a number of internal functions that apply directly to the controller. When an error is detected, the controller gives a system alarm, and shows one of the following messages.



Always contact your installer when a system alarm occurs!

Description	Type	Cause
Options corrupted.	Alarm	The system setup no longer correct.
Settings corrupted .	Alarm	The system settings are no longer correct.
Minimum/maximum value corrupted.	Warning	The stored minimum and maximum values are no longer correct.
I ₂ C error.	Alarm	Internal communication problem in the Orion-MC.
EEPROM error.	Alarm	Internal communication problem with the EEPROM settings and options.
RTC error.	Warning	Problem with the real-time clock (program flow).
TIMER error.	Warning	The internal clock does not operate correctly.
HTRAP error.	Warning	The internal memory does not operate correctly.

Description	Type	Cause
PLL error.	Warning	The PLL circuit does not operate correctly.
IO board ADC error.	Warning	The analog/digital converter does not operate correctly.

7.2 Diagnostics

Introduction

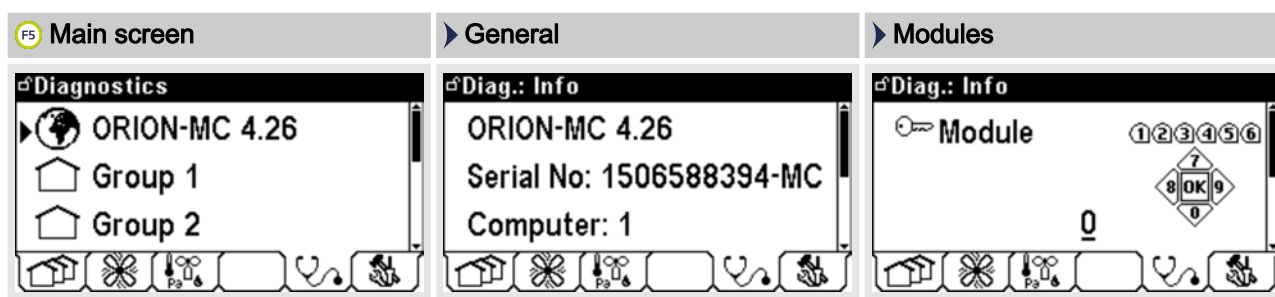
Personnel: ☒ User



Control level: „F5 → Diagnostics“

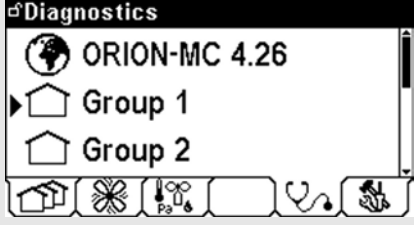
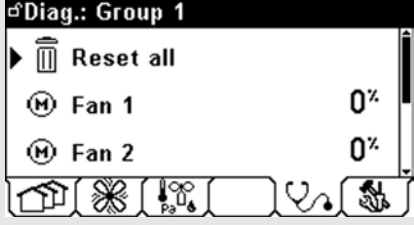
The diagnostics section gives detailed information on the operation of all the controls, for example:

- Important control values
- Minimum and maximum values of measurements
- The status of the inputs and outputs
- The assigned inputs and outputs



At „Module“, enter a licence key to enable optional features such as CANopen, ventilation or air mixing units.

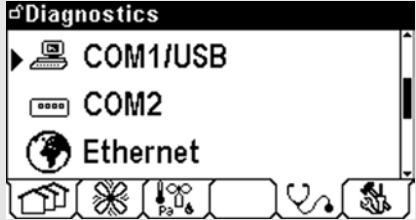
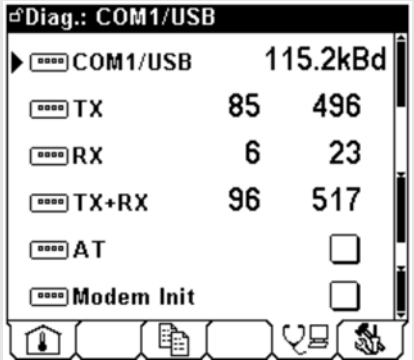
Ventilation Control

F5 Main screen	General	Details
 Diagnosics ORION-MC 4.26 Group 1 Group 2	 Diag.: Group 1 Reset all Fan 1 0% Fan 2 0%	 Diag.: Fan 1 Running hours 0h Power 0W Alarm Code 0000h Temp motor 0°C Temp electronics 0°C Temp power module 0°C Maximum rpm 0
„ORION-MC xx“: The software version of this device	„Running time“: Running time of the virtual motor group „Fan x“: Current position of this motor	„Running hours“: The number of hours this fan has been in operation. „Power“: The power consumption (information from the fan). „Alarm code“: The code of the alarm message (information from the fan). „Temp“: The measured temperatures (information from the fan). „Maximum rpm“: The maximum speed (information from the fan).

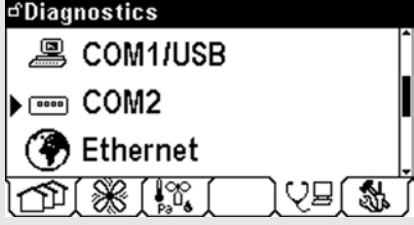
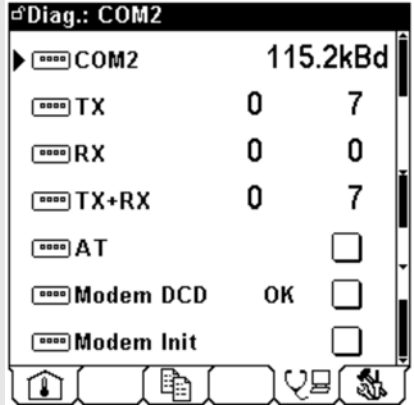


- It is possible that the diagnostics information is not displayed when the fan:
 - does not send information, or
 - sends other information
- It is possible that the Orion-MC receives different alarm messages when the system includes a large number of fans. After you solved the error, you can delete the alarm messages with the „Reset all“ function.

COM1/USB

F5 Main screen	General	Details
	 In case of a box, you can press ► to enable or disable the option.	No details
	„COM1/USB“ to „TX+RX“ gives communication information. „AT“: Send an AT command to the modem. „Modem Init“: Send an initialization string to the modem.	

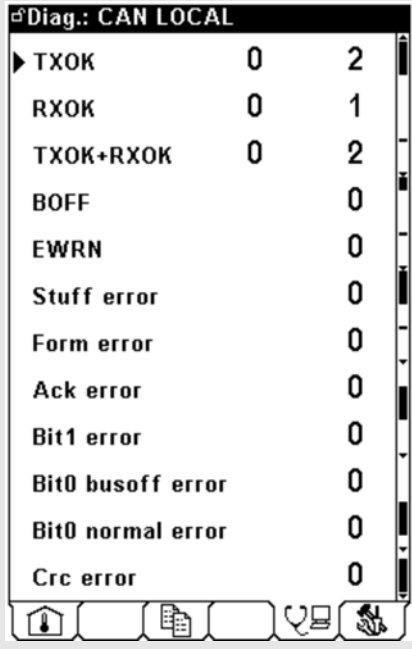
COM2

F5 Main screen	General	Details																		
 Diagnosics - COM1/USB - COM2 - Ethernet	 Diag.: COM2 COM2 115.2kBd <table border="1"> <tr> <td>TX</td> <td>0</td> <td>7</td> </tr> <tr> <td>RX</td> <td>0</td> <td>0</td> </tr> <tr> <td>TX+RX</td> <td>0</td> <td>7</td> </tr> <tr> <td>AT</td> <td></td> <td>☐</td> </tr> <tr> <td>Modem DCD</td> <td>OK</td> <td>☐</td> </tr> <tr> <td>Modem Init</td> <td></td> <td>☐</td> </tr> </table> In case of a box, you can press ► to enable or disable the option.	TX	0	7	RX	0	0	TX+RX	0	7	AT		☐	Modem DCD	OK	☐	Modem Init		☐	No details
TX	0	7																		
RX	0	0																		
TX+RX	0	7																		
AT		☐																		
Modem DCD	OK	☐																		
Modem Init		☐																		
	„COM1/USB“ to „TX+RX“ gives communication information. „AT“: Send an AT command to the modem. „Modem DCD“: Test the DCD signal. „Modem Init“: Send an initialization string to the modem.																			

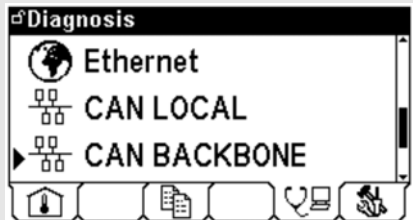
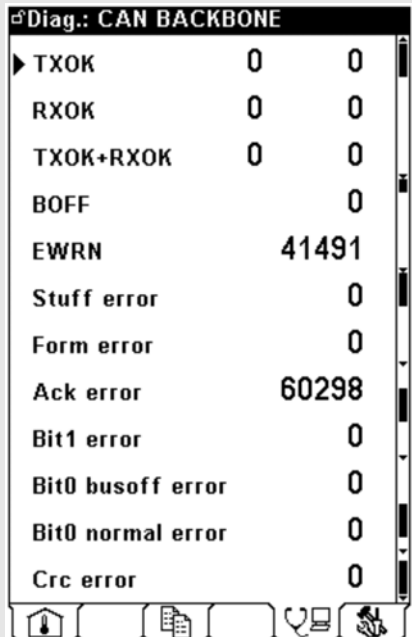
Ethernet

F5 Main screen	General	Details
Diagnosics COM1/USB COM2 Ethernet 	Diag.: Ethernet Port 5843 IP 0.0.0.0 Mask 0.0.0.0 Gate 0.0.0.0 MAC 00:0B:3C:33:04:68 PC 1 NO CONNECTION PC 2 NO CONNECTION PC 3 NO CONNECTION TX 0 0 RX 0 0 RTX 0 NOACK 0 CONNECT 0 0 Reset Ethernet ☐ 	No details

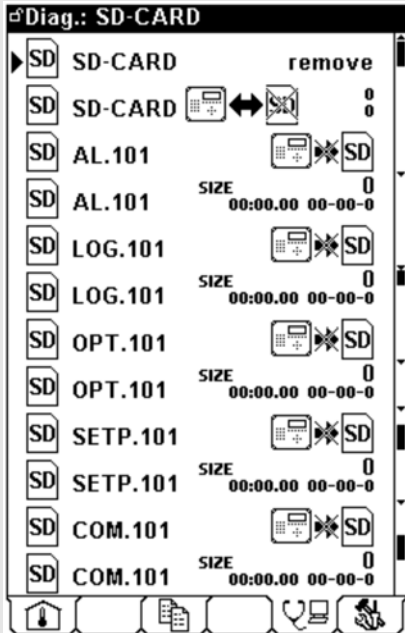
CAN Local

F5 Main screen	General	Details																																				
 Diagnosics - Ethernet - CAN LOCAL - SD-CARD	 Diag.: CAN LOCAL <table> <tr><td>TXOK</td><td>0</td><td>2</td></tr> <tr><td>RXOK</td><td>0</td><td>1</td></tr> <tr><td>TXOK+RXOK</td><td>0</td><td>2</td></tr> <tr><td>BOFF</td><td></td><td>0</td></tr> <tr><td>EWRN</td><td></td><td>0</td></tr> <tr><td>Stuff error</td><td></td><td>0</td></tr> <tr><td>Form error</td><td></td><td>0</td></tr> <tr><td>Ack error</td><td></td><td>0</td></tr> <tr><td>Bit1 error</td><td></td><td>0</td></tr> <tr><td>Bit0 busoff error</td><td></td><td>0</td></tr> <tr><td>Bit0 normal error</td><td></td><td>0</td></tr> <tr><td>Crc error</td><td></td><td>0</td></tr> </table>	TXOK	0	2	RXOK	0	1	TXOK+RXOK	0	2	BOFF		0	EWRN		0	Stuff error		0	Form error		0	Ack error		0	Bit1 error		0	Bit0 busoff error		0	Bit0 normal error		0	Crc error		0	No details
TXOK	0	2																																				
RXOK	0	1																																				
TXOK+RXOK	0	2																																				
BOFF		0																																				
EWRN		0																																				
Stuff error		0																																				
Form error		0																																				
Ack error		0																																				
Bit1 error		0																																				
Bit0 busoff error		0																																				
Bit0 normal error		0																																				
Crc error		0																																				
	Communication information																																					

CAN Backbone (CANopen)

F5 Main screen	General	Details																																				
 Diagnosis - Ethernet - CAN LOCAL - CAN BACKBONE	 Diag.: CAN BACKBONE <table> <tr><td>TXOK</td><td>0</td><td>0</td></tr> <tr><td>RXOK</td><td>0</td><td>0</td></tr> <tr><td>TXOK+RXOK</td><td>0</td><td>0</td></tr> <tr><td>BOFF</td><td></td><td>0</td></tr> <tr><td>EWRN</td><td></td><td>41491</td></tr> <tr><td>Stuff error</td><td></td><td>0</td></tr> <tr><td>Form error</td><td></td><td>0</td></tr> <tr><td>Ack error</td><td></td><td>60298</td></tr> <tr><td>Bit1 error</td><td></td><td>0</td></tr> <tr><td>Bit0 busoff error</td><td></td><td>0</td></tr> <tr><td>Bit0 normal error</td><td></td><td>0</td></tr> <tr><td>Crc error</td><td></td><td>0</td></tr> </table>	TXOK	0	0	RXOK	0	0	TXOK+RXOK	0	0	BOFF		0	EWRN		41491	Stuff error		0	Form error		0	Ack error		60298	Bit1 error		0	Bit0 busoff error		0	Bit0 normal error		0	Crc error		0	No details
TXOK	0	0																																				
RXOK	0	0																																				
TXOK+RXOK	0	0																																				
BOFF		0																																				
EWRN		41491																																				
Stuff error		0																																				
Form error		0																																				
Ack error		60298																																				
Bit1 error		0																																				
Bit0 busoff error		0																																				
Bit0 normal error		0																																				
Crc error		0																																				

SD Card

F5 Main screen	General	Details
	 Some options have additional choices. Select an option and press ► to access it. „SD-CARD“: The status of the SD card. „AL “ to „COM“: The files on the SD card.	No details

Change the Status of the SD card

Press ► when the cursor is on „SD CARD“. Change the status with ▲▼. Confirm with **OK**. You can give the SD card these statuses:

„Remove“	You can safely remove the SD card from the Orion-MC.
„Ask remove“	Data is still being written to the card. You cannot yet remove the SD card.
„Current“	You can use the SD card to save or restore data.

Change the Status of the File

Press ► when the cursor is on the file. Change the status with ▲▼. Confirm with **OK**. You can give the SD card these statuses:

	No action occurs.
	The data are stored on the SD card. With log data the data are supplemented continually and the ➡ symbol is always displayed. With one-off data (such as options) the symbol changes back to ➡ once the data has been saved.
	The data are read out from the SD card and send to the Orion-MC (options, for example). Once the read-out has been completed, the symbol changes back to .

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